HAMILTON CYCLES IN 1-TOUGH GRAPHS WITH NO INDUCED $P_4 \cup P_1$

QIUYU CHEN

Abstract. We prove that every finite simple 1-tough graph of order at least three with no induced disjoint union of a four-vertex path and an isolated vertex has a Hamilton cycle. The proof first obtains a spanning 2-factor from a barrier argument. It then reduces a minimum-cycle factor to canonical groups and finds a cut in the original graph that contradicts toughness.

1. Introduction

Background. Toughness compares the size of a vertex cut with the number of components it leaves. A Hamilton cycle imposes the necessary condition that every separating set has at least as many vertices as the remaining graph has components. For graphs excluding small induced subgraphs, this condition can also be sufficient. Nikoghosyan proved the sufficiency of 1-toughness when an induced $P_3 \cup P_1$ is forbidden, and conjectured the corresponding statement for $P_4 \cup P_1$ in [Nik13, Conjecture 7].

Recent results address the same forbidden configuration with stronger toughness assumptions. Shan proved Hamiltonicity for 23-tough graphs in [Shan26]; Lu, Wu, and Zhou reduced the sufficient bound to 7 in [LWZ26, Theorem 1.1]. We prove the statement with toughness 1. The argument combines the factor criterion of Tutte [Tut54, Theorem C], in the notation of [KS19, Lemma 3.1], with cograph path covers and the coloured-cycle separator theorem of [Yeo97].

1.1. The Hamiltonicity theorem.

For a graph G, let $c(G)$ denote its number of components. A graph is *1-tough* if

$$|S| \geq c(G - S) \qquad \text{whenever } c(G - S) \geq 2.$$

Every cycle in this paper is a simple cycle of order at least three.

Theorem 1.1 (Hamilton cycles)**.** *Let G be a finite simple 1-tough graph of order at least three. If G has no induced subgraph isomorphic to $P_4 \cup P_1$, then G has a Hamilton cycle consisting entirely of edges of G.*

The proof does not assume that a 2-factor exists. Theorem 3.1 establishes its existence, and Theorem 13.2 excludes every non-Hamiltonian minimum-cycle factor with at least two cycles.

2020 *Mathematics Subject Classification.* 05C45, 05C38.

Key words and phrases. Hamilton cycle, toughness, forbidden induced subgraph, 2-factor.

Proof strategy. First, a biased barrier in the 2-factor criterion would give a vertex cut violating 1-toughness (Theorem 3.1). Choose a factor with the fewest cycles. A separate argument handles factors all of whose cycle supports induce cographs: the coloured separator reduces the host to two sides, and a terminal path-cover spectrum closes a spanning cycle using their existing edges (Theorem 7.1).

Otherwise, the factor has a unique noncograph block H. Every remaining block attaches to H through a canonical tight set, called its gate. Blocks with different gates are anticomplete. The proof compares minimum path covers before and after foreign cycles are added. A strict saving or a newly eligible old endpoint can be witnessed by a single added cycle (Theorem 9.1); protected path constructions then turn that event into a merger of factor cycles.

These merger obstructions yield a cut containing all attachments of one gate group. The pivotal final step treats vertices in different components of that group: an attachment escaping the cut would exhibit an induced $P_4 \cup P_1$. The whole group is consequently separated by a cut too small for the resulting component count.

Organization. Section 2 fixes conventions, and Section 3 proves factor existence. Sections 4 and 5 develop cograph endpoints and protected paths. Sections 6 and 7 settle the all-cograph case. Sections 8 and 9 prove absorption and locality. Sections 10–13 obtain canonical groups and the final cut, followed by the proof of Theorem 1.1.

2. Preliminaries

All graphs are finite, simple, and undirected. For $X \subseteq V(G)$, the notation $G[X]$ denotes the induced subgraph and $G - X = G[V(G) \setminus X]$. We write $N_H(x) = N_G(x) \cap V(H)$, even when $x \notin V(H)$. The order of a graph is its number of vertices. A graph is traceable if it has a spanning path. A 2-factor is a partition of the vertex set into simple cycles, each of order at least three; its blocks are the induced graphs on the specified cycle supports.

Every named union, such as $H + D$, is induced with all its existing ambient edges. A complete join is denoted by $X \vee Y$ and is used only when every edge between its two disjoint sides is present. Component counts include $c(\varnothing) = 0$. We identify induced graphs with their vertex sets when writing deletions, intersections and cardinalities.

3. Existence of a 2-factor

We use the factor criterion to rule out a negative barrier by an actual toughness cut.

For disjoint $S, T \subseteq V(G)$, let $h(S,T)$ count the components D of $G - (S \cup T)$ for which the number $e(D,T)$ of edges from D to T is odd, and put

$$\Delta(S,T) = 2|S| - 2|T| + \sum_{t \in T} d_{G-S}(t) - h(S,T).$$

Tutte's criterion [Tut54, Theorem C], in the form recorded in [KS19, Lemma 3.1], says that G has a 2-factor if and only if $\Delta(S,T) \geq 0$ for every disjoint pair. The parity used below follows from a separate count.

Theorem 3.1 (Spanning 2-factor)**.** *Every finite simple 1-tough graph of order at least three with no induced $P_4 \cup P_1$ has a 2-factor.*

Proof. We first compute $\Delta(S,T)$ for an arbitrary disjoint pair, with no toughness or forbidden-subgraph assumption. Put $U = V(G) \setminus (S \cup T)$. For each component D of $G[U]$, write

$$m_D = e(D,T) = 2q_D + \epsilon_D, \qquad \epsilon_D \in \{0,1\}.$$

Counting an edge of $G[T]$ twice and an edge from T to U once gives

$$\sum_{t\in T} d_{G-S}(t) = 2e(T) + \sum_D m_D, \qquad h(S,T) = \sum_D \epsilon_D.$$

Consequently

$$\Delta(S,T) = 2\left(|S| - |T| + e(T) + \sum_D q_D\right). \tag{1}$$

The calculation includes empty T and empty U, with empty sums equal to zero. Thus every value of Δ is even. A negative value is at most -2, while a pair with no negative value has $\Delta \geq 0$.

Suppose that G has no 2-factor. Choose a pair (S,T) with $\Delta(S,T) \leq -2$, first maximizing $|S|$ and then minimizing $|T|$. Let $U = V(G) \setminus (S \cup T)$. For $t \in T$, let h_t count the odd components of $G[U]$ meeting t. Removing t from T leaves the same S and a smaller T, so the new pair is not a negative barrier. Odd components not meeting t remain odd. Hence

$$\begin{aligned} 0 &\leq \Delta(S, T \setminus \{t\}) \\ &= \Delta(S,T) + 2 - d_{G-S}(t) + h(S,T) - h(S, T\setminus\{t\}) \\ &\leq -d_{G-S}(t) + h_t. \end{aligned} \tag{2}$$

Each component counted by h_t contributes at least one edge at t, so $d_{G-S}(t) \geq h_t$. Equality holds throughout (2). It follows that T is independent, no even component of $G[U]$ meets T, and each $t \in T$ has exactly one neighbor in every odd component that it meets.

Let D be an odd component and $x \in V(D)$. Moving x into S increases $|S|$, so the resulting pair is not a negative barrier. Every other odd component persists, giving $h(S \cup \{x\}, T) \geq h(S,T) - 1$. Therefore

$$\begin{aligned} 0 &\leq \Delta(S \cup \{x\}, T) \\ &= \Delta(S,T) + 2 - e(x,T) + h(S,T) - h(S \cup \{x\}, T) \\ &\leq 1 - e(x,T). \end{aligned}$$

Thus x has at most one neighbor in T. Together with the restriction at vertices of T, the edges between D and T form a matching.

We show that this matching has at most two edges. Suppose it has distinct edges $t_i d_i$, $i = 1,2,3$. Any two attachment vertices have distance at most two in $G[D]$. If a shortest d_1-d_2 path had length at least three, its first three vertices, preceded by t_1, would induce a P_4. The vertex t_2 would be isolated from those four vertices: its sole neighbor in D is d_2, and T is independent. This would induce $P_4 \cup P_1$.

No two attachment vertices are adjacent. If $d_1 d_2$ were an edge, t_1, d_1, d_2, t_2 would induce a P_4 isolated from t_3. Thus d_1, d_2, d_3 are pairwise nonadjacent. Take a length-two path d_1, w, d_2, whose existence follows from the distance bound. Then t_1, d_1, w, d_2 induce a P_4. To keep t_3 from being isolated from it, the matching

condition forces $w = d_3$. But d_3 would then be adjacent to both d_1 and d_2, a contradiction.

Every odd component has a positive odd number of edges to T, so it has exactly one. Even components have none, and T is independent. Hence

$$\sum_{t\in T} d_{G-S}(t) = h(S,T), \qquad \Delta(S,T) = 2|S| - 2|T| \le -2.$$

In particular $|T| > |S|$. In the original graph $G - S$, a component of $G[U]$ attaches to at most one vertex of T. No two vertices of T can therefore lie in the same component of $G - S$, and

$$c(G - S) \ge |T| > |S|.$$

When $|T| \ge 2$, this contradicts 1-toughness.

The remaining case has $T = \{t\}$ and $S = \varnothing$. A 1-tough graph of order at least three is connected: a disconnected graph violates toughness at the empty cut. It also has minimum degree at least two. A degree-zero vertex contradicts connectedness, while deleting the sole neighbor of a degree-one vertex leaves that vertex and at least one other vertex in separate components. Each component of $G - t$ has at most one edge to t. Since $d(t) \ge 2$, the graph $G - t$ has at least two components. The original cut $\{t\}$ again violates 1-toughness. Thus no negative barrier exists, and the cited criterion gives a 2-factor. □

4. Cograph endpoint cuts

The cotree separates a cograph into disjoint unions and complete joins. We use those two operations to track minimum path covers, their possible endpoints, and the cuts on which the path-cover bound is attained.

Definition 4.1. All graphs are finite, simple, and undirected. A path cover of a graph K consists of vertex-disjoint nonempty paths partitioning $V(K)$; singleton paths are allowed. Write $p(K)$ for its minimum size and $c(K)$ for the number of components, with $p(\emptyset) = c(\emptyset) = 0$. A cycle is a simple cycle of order at least three.

A vertex is *eligible* in K if it is an endpoint of a path in some minimum path cover. Write $E(K)$ for the eligible vertices and $F(K) = V(K) \setminus E(K)$. Eligibility does not prescribe endpoints simultaneously. For nonempty K, let $D(K)$ be the graph on $E(K)$ in which distinct vertices are adjacent precisely when they end distinct paths in one minimum cover. Thus $D(K)$ is edgeless when $p(K) = 1$, with one independent part. It is not defined for the empty graph.

For a connected traceable graph H of order at least two, let $R(H)$ be the graph on $E(H)$ in which distinct u, v are adjacent precisely when H has a spanning u-v path. A graph is *1-tough* if $|S| \ge c(H - S)$ whenever $c(H - S) \ge 2$.

Lemma 4.2 (Cotree decomposition)**.** *A cograph is a P_4-free graph. Its induced subgraphs are cographs, and every connected cograph of order at least two is a nontrivial complete join of nonempty smaller cographs. Conversely, disjoint union and complete join preserve P_4-freeness. A connected cograph has diameter at most two.*

In a graph with no induced $P_4 \cup P_1$, every induced P_4 dominates every other vertex. Consequently every vertex antineighborhood is a cograph, as is every component of a disconnected induced subgraph.

Proof. Fix a connected P_4-free graph K of order at least two and a vertex v. If v is universal, it supplies one side of a nontrivial join. Otherwise let B_i be the components of the nonneighbors of v, and put $A = N_K(v)$. A vertex of A that meets B_i is complete to B_i: an edge across the boundary of its neighborhood in B_i would form an induced path v, a, b, b'. Let A_i be the vertices of A complete to B_i. Connectedness makes each A_i nonempty.

A vertex $a \in A_i$ is adjacent to every vertex $b \in A \setminus A_i$, since otherwise B_i, a, v, b contains an induced P_4. The sets A_i are nested: if $a \in A_i \setminus A_j$ and $b \in A_j \setminus A_i$, then a vertex of B_i, followed by a, b, and a vertex of B_j form an induced P_4. A smallest A_i is therefore complete to its complement in K, giving the join. Recursion gives the cotree. Induced subgraphs remain P_4-free, and both disjoint union and complete join preserve this property because P_4 and its complement are connected. A shortest path of length at least three contains an induced P_4, proving the diameter assertion.

An induced P_4 with a vertex adjacent to none of its vertices would give an induced $P_4 \cup P_1$. A P_4 in a vertex antineighborhood has just such a vertex. A P_4 in one component of a disconnected induced subgraph has one in another component. □

Proposition 4.3 (Join recurrence and balanced joins)**.** *For a disjoint union $K = \bigcup_i K_i$, we have $p(K) = \sum_i p(K_i)$. For a nontrivial complete join of nonempty cographs K_i, put $r_i = |V(K) \setminus V(K_i)|$. Then*

$$p(K) = \max\bigl(1, \max_i \{p(K_i) - r_i\}\bigr). \tag{1}$$

At most one child satisfies $p(K_i) - r_i \geq 1$. If $p(K_i) \leq r_i$ for every child, then K is Hamiltonian when its order is at least three. At order two it is K_2, which has a spanning path but no cycle.

Proof. The union formula follows because a cover path stays in one component. For the join, deleting the r_i outside vertices from any q-path cover leaves at most $q + r_i$ paths covering K_i. This gives the lower bound in (1). If two children i, j both had positive differences, then $p(K_i) \leq |K_i|$ would give $|K_i| > r_i \geq |K_j|$ and the reverse inequality, which is impossible.

Suppose a child X satisfies $p(X) = r + q \geq r + 1$, where r is its number of outside vertices. Take a minimum cover of X, arrange its $r + q$ paths into q nonempty chains, and use each of the r outside vertices once between consecutive paths. Every connecting edge is an original join edge. This gives a q-path cover.

It remains to treat $p(K_i) \leq r_i$ for all i. Start with $t_i = p(K_i)$. If one count $p(K_b)$ exceeds the sum of the others, increase the other counts until their sum is $p(K_b)$. Their total capacity is $r_b \geq p(K_b)$, and every resulting count is at most the sum of the others. If no count is excessive, keep the initial counts. In either case,

$$p(K_i) \leq t_i \leq |K_i|, \qquad t_i \leq \sum_{j \neq i} t_j \quad \text{for every } i.$$

Cutting $t_i - p(K_i)$ edges of a fixed minimum cover of K_i realizes t_i paths.

Relabel a largest count as $t_1 = M$, and place its M path tokens on a circle. Number the following gaps $0, \ldots, M - 1$. Make one list of every other token, in consecutive color batches, and put token ℓ in gap ℓ mod M. The gap index continues across color changes. Each other color occupies at most M consecutive list positions, so it occurs at most once in each gap. The list has at least M tokens, so every gap is nonempty. Reading around the circle therefore places different colors on either

side of every seam, including the closing seam. For example, counts $(3, 2, 2)$ give the cyclic color word $1, 2, 3, 1, 2, 1, 3$.

Replace each token by its fixed path, with a fixed orientation. Internal path edges and all seams are original edges of K. With at least three path blocks this is a simple spanning cycle. With exactly two blocks and at least three vertices, one block is non-singleton; its two endpoints make the two seams distinct edges, again giving a simple spanning cycle. The remaining order-two graph is K_2. Thus the balanced join has $p(K) = 1$, completing (1). □

Proposition 4.4 (Saturated joins)**.** *Suppose a join K has $p(K) = q \geq 1$ and a child X with $p(X) = q + r$, where $Y = V(K) \setminus V(X)$ and $r = |Y|$. Every minimum q-cover of K starts and ends each path in X, uses each vertex of Y as an internal singleton connector, and has no consecutive vertices of Y. Deleting Y leaves a minimum $(q + r)$-cover of X. Moreover,*

$$E(K) = E(X), \qquad F(K) = Y \cup F(X).$$

If $q \geq 2$, then $D(K) = D(X)$. No such identity for $D(K)$ is asserted when $q = 1$.

Proof. Deleting Y from a minimum q-cover leaves at most $q + r$ nonempty X-blocks. These blocks cover X, whose path-cover number is $q + r$, so equality holds. A path containing s vertices of Y contributes at most $s + 1$ nonempty X-blocks. Equality in the sum forces each path to begin and end in X, each Y-vertex to lie internally, and each pair of successive Y-vertices to have an X-block between them. Thus the X-blocks form a minimum cover.

Every endpoint of a minimum cover of K is consequently eligible in X. Conversely, start with a minimum cover of X realizing a chosen eligible endpoint, and chain its $q + r$ paths into q paths using the r vertices of Y as connectors. Put the chosen endpoint on the exterior of its chain. This proves the assertions about E and F. If $q \geq 2$, start instead with one cover of X realizing two vertices as endpoints of distinct paths. Put those paths in different final chains. This realizes the same pair in $D(K)$; deletion of Y gives the reverse implication. When $q = 1$, $D(K)$ is edgeless by definition. □

Proposition 4.5 (The endpoint cut and distinct-path relation)**.** *For every cograph K, including the empty graph,*

$$c(K - F(K)) = |F(K)| + p(K). \tag{2}$$

If K is nonempty, $D(K)$ is complete multipartite on $E(K)$. For every independent part P of $D(K)$, there is $S \subseteq F(K)$ such that $c(K - S) = |S| + p(K)$ and $(F(K) \cup P) \setminus S$ lies in one component of $K - S$.

Proof. We induct along the cotree. The empty graph satisfies (2). For a singleton, $E(K) = V(K)$, $F(K) = \emptyset$, and its edgeless $D(K)$ has one part, witnessed by $S = \emptyset$.

For a disjoint union, minimum covers split into minimum covers of the children. Thus E and F are the respective unions, and (2) adds over the children. Endpoint pairs from different children occur on distinct paths of a common minimum cover. Hence $D(K)$ is the join of the child relations and is complete multipartite. If a part P comes from child K_i with witness S_i, take $S = S_i \cup \bigcup_{j \neq i} F(K_j)$. The child cut identities add to give the required equality, while the remaining vertices of $F(K) \cup P$ lie in the component certified in $K_i - S_i$.

Now suppose K is connected and $q = p(K) \geq 2$. Proposition 4.3 gives a unique child X with $p(X) = q + r$, where Y is its outside set of size r. Proposition 4.4 gives $E(K) = E(X)$, $F(K) = Y \cup F(X)$, and $D(K) = D(X)$. Since $K - F(K) = X - F(X)$, the inductive cut identity for X gives (2) for K. For a part P inherited from X, extend its inductive witness S_X to $S = Y \cup S_X$. The same component contains $(F(K) \cup P) \setminus S$, and $c(K - S) = |S_X| + p(X) = |S| + q$.

Finally suppose K is connected with $p(K) = 1$. Its graph $D(K)$ is edgeless, with sole part $E(K)$; $S = \emptyset$ witnesses the part assertion. If no join child has $p(K_i) = r_i + 1$, Proposition 4.3 gives a Hamilton cycle, or the single edge of K_2. Deleting an incident cycle edge, or using that single edge, shows that every vertex is eligible. Hence $F(K) = \emptyset$ and (2) holds. Otherwise the unique child X with $p(X) = r + 1$ satisfies the conclusion of Proposition 4.4. The inductive identity for X then gives (2) for K. □

Lemma 4.6 (Attaining cuts)**.** *Let K be any graph. If $X \subseteq V(K)$ satisfies $c(K - X) = |X| + p(K)$, then $X \subseteq F(K)$. For a cograph, $F(K)$ is the unique largest set satisfying this equality.*

Proof. Choose a minimum cover with $p(K)$ paths. If a path contains s deleted vertices, its surviving vertices form at most $s + 1$ nonempty consecutive pieces; deleting the whole path can leave none. Summing over the cover and then adding the other edges of $K - X$ gives

$$c(K - X) - |X| \leq p(K),$$

also when $X = \emptyset$ or $X = V(K)$. If X contains an eligible vertex v, choose a minimum cover with v at an endpoint and delete v first. This deletion does not increase the number of path pieces. Each of the other $|X| - 1$ deletions increases it by at most one, so $c(K - X) \leq p(K) + |X| - 1$, contradicting attainment. Thus $X \subseteq F(K)$. Proposition 4.5 says that $F(K)$ itself attains equality for a cograph, and every other attaining set is contained in it. □

Proposition 4.7 (Restriction to cover paths)**.** *Let K be a nonempty cograph, and fix any minimum path cover $\mathcal{P}$. If L is the subgraph induced by the vertices of any selected subcollection of m paths of $\mathcal{P}$, then $p(L) = m$ and $E(L) = E(K) \cap V(L)$. In particular this holds when L is induced by one path, including a singleton path.*

Proof. A cover of L with fewer than m paths could replace the selected paths in $\mathcal{P}$, so $p(L) = m$. Put $F = F(K)$ and $F_0 = F \cap V(L)$. Deleting F from the paths of $\mathcal{P}$ gives at most $|F| + p(K)$ nonempty segments. Equality in (2) forces every segment to be a whole component of $K - F$, with no edge between distinct segments. In particular, F avoids the cover endpoints, no two vertices of F are consecutive on a cover path, and $c(L - F_0) = |F_0| + m$.

A minimum cover of L can replace the selected paths of $\mathcal{P}$. Hence every locally eligible vertex is globally eligible, so $F_0 \subseteq F(L)$. Suppose this inclusion were strict. Replace F_0 in the global cut F by $F(L)$. The components of $K - F$ meeting L lie wholly inside L, so the other components do not change. Proposition 4.5 gives $c(L - F(L)) = |F(L)| + m$, compared with $c(L - F_0) = |F_0| + m$. The replacement cut therefore still satisfies $c(K - S) = |S| + p(K)$. It contains a vertex of $F(L) \setminus F_0$, which is eligible in K, contradicting Lemma 4.6. Thus $F(L) = F_0$, as required. □

Proposition 4.8 (Hamilton-path endpoint classes)**.** *Let H be a connected traceable cograph of order at least two. Then $R(H)$ is complete multipartite on $E(H)$, and precisely one of the following cases holds.*

1. *If a join child X satisfies $p(X) = r + 1$, where $Y = V(H) \setminus V(X)$ and $r = |Y|$, then $E(H) = E(X)$, $F(H) = Y \cup F(X)$, and $R(H) = D(X)$. For every independent part P of $R(H)$, there is a nonempty $S \subseteq F(H)$ such that $c(H - S) = |S| + 1$ and $(F(H) \cup P) \setminus S$ lies in one component of $H - S$.*
2. *If no join child satisfies $p(X) = r + 1$, then $E(H) = V(H)$, and every independent part P of $R(H)$ satisfies $c(H - P) = |P|$.*

Proof. In the first case, Proposition 4.4 gives the assertions about E and F. Deleting Y from a spanning path leaves a minimum cover of X whose two outer endpoints belong to distinct paths. Conversely, orient those two paths first and last in a minimum cover of X, and chain all $r + 1$ paths with the r singleton connectors in Y. All seams are original join edges. Thus $R(H) = D(X)$. For a part P, take its witness $S_X \subseteq F(X)$ from Proposition 4.5 and set $S = Y \cup S_X$. This set is nonempty, and the cut equality and component containment follow from the corresponding assertions for $X - S_X$.

We prove the second case by induction on $|H|$, using the first case for any smaller critical graph. At order two, $H = K_2$; its two eligible vertices form singleton parts of $R(H)$, and deleting either part leaves one component. For larger H, the absence of a critical child and Proposition 4.3 give an original Hamilton cycle C. If $\emptyset \neq S \subsetneq V(H)$, deleting S from C leaves consecutive path segments. Each segment has a distinct predecessor in S on C, so there are at most $|S|$ segments. Additional edges can only merge them, and hence $c(H - S) \leq |S|$. The same inequality is immediate for $S = V(H)$. The empty set leaves H connected. In particular, H is 1-tough. Deleting either incident cycle edge at any vertex gives a spanning path ending there. As $p(H) = 1$, every vertex is eligible.

Fix $x \in V(H)$ and put $K = H - x$. Deleting x from C shows that K is connected and traceable. For $y \in V(K)$, we claim

$$H \text{ has a spanning } x\text{-}y \text{ path} \iff y \in E(K). \tag{3}$$

Deleting x proves necessity. For sufficiency, suppose $y \in E(K)$ but no such path exists. By induction $R(K)$ is complete multipartite; let P be the part containing y. A neighbor of x in $E(K) \setminus P$ could be joined to y by a spanning path of K, then extended through its original edge to x. Thus $N_H(x) \subseteq F(K) \cup P$.

If K is critical, its part certificate supplies a nonempty $S \subseteq F(K)$ with $c(K - S) = |S| + 1$, while all neighbors of x outside S lie in one component of $K - S$. Adding x can join no other components, so $c(H - S) \geq |S| + 1 \geq 2$, contrary to 1-toughness. If K is balanced, then $F(K) = \emptyset$ and $c(K - P) = |P|$. The vertex x is isolated in $H - P$, giving $c(H - P) = |P| + 1 \geq 2$, again a contradiction. This proves (3).

Set $Q_x = \{x\} \cup F(H - x)$. Proposition 4.5 gives

$$c(H - Q_x) = |F(H - x)| + 1 = |Q_x|.$$

Two distinct vertices of Q_x cannot be endpoints of a spanning path: deleting a set of s vertices containing both endpoints from such a path leaves at most $s - 1$ nonempty pieces, whereas this cut leaves s components. By (3), Q_x consists exactly of x and the vertices incompatible with it in $R(H)$. If $y \in Q_x$, the same cut shows that y is incompatible with every other vertex of Q_x; hence $Q_x \subseteq Q_y$. Interchanging x and y

gives equality. The sets Q_x therefore partition $V(H)$ into the independent parts of a complete multipartite $R(H)$, and the displayed cut equality gives $c(H-P) = |P|$ for each part. □

Proposition 4.9 (Profile scattering maximum)**.** *Let Q be any graph and $A \subseteq V(Q)$ such that $K = Q - A$ is a nonempty cograph. Put $\delta_A = p(K) - |A|$ and $T_A = A \cup F(K)$. Among all sets $T \supseteq A$, including those for which $Q - T$ has fewer than two components, the maximum of $c(Q-T) - |T|$ is δ_A. The set T_A is the unique largest maximizer. No Hamiltonicity or toughness assumption on Q is needed.*

Proof. Write $T = A \cup X$ uniquely with $X \subseteq V(K)$. Then $Q - T = K - X$ and $|T| = |A| + |X|$. The path-cover comparison in Lemma 4.6 gives

$$c(Q-T) - |T| = c(K-X) - |X| - |A| \leq p(K) - |A| = \delta_A.$$

This includes $A = \emptyset$, $X = \emptyset$, and $X = V(K)$. Taking $X = F(K)$ attains equality by Proposition 4.5. Any other equality case has $X \subseteq F(K)$ by Lemma 4.6, so every maximizer is contained in T_A. Proper subsets may also attain the maximum. □

5. Rows, paths, and canonical gates

This section relates tight cuts to spanning-path endpoints and proves a path construction that preserves specified cycle edges.

For a Hamiltonian graph H, call $T \subseteq V(H)$ *tight* when $c(H-T) = |T|$. A profile of x into H is its actual neighborhood $N_H(x)$. Let $R(H)$ be the graph whose distinct adjacent pairs are the endpoint pairs of spanning paths of H.

Proposition 5.1 (Canonical saturated container)**.** *Let H be Hamiltonian of order at least three, and let A be a nonempty vertex set such that $K = H - A$ is a cograph. A tight set containing A exists if and only if $p(K) = |A|$. In that case*

$$S_A = A \cup F(K)$$

is a proper tight set, and every tight set containing A is contained in S_A. Thus S_A is the unique largest tight container of A. We call A saturated *when $p(K) = |A|$.*

Proof. Deleting A from a Hamilton cycle gives $p(K) \leq |A|$. If $A = V(H)$, then $p(K) = 0 < |A|$, while $V(H)$ is not tight. Assume henceforth that K is nonempty.

Let T be tight and contain A, and put $X = T \setminus A$. Tightness gives $c(K-X) = |A| + |X|$. Delete X from one minimum $p(K)$-path cover of K. Each deleted vertex increases the number of nonempty path pieces by at most one, and those pieces cover $K - X$. Consequently

$$|A| + |X| = c(K-X) \leq p(K) + |X|.$$

Together with $p(K) \leq |A|$, this forces $p(K) = |A|$. It also shows, only after equality is established, that X attains the cut value $p(K) + |X|$. The preceding maximality result for $F(K)$ gives $X \subseteq F(K)$.

Conversely, if $p(K) = |A|$, the cut identity gives $c(H - S_A) = c(K - F(K)) = p(K) + |F(K)| = |S_A|$. The preceding paragraph places every other tight container inside S_A. Finally, $K - F(K)$ has $p(K) + |F(K)| \geq 1$ components, so S_A is proper. □

Proposition 5.2 (Exact spanning-path endpoints)**.** *Under the hypotheses of Proposition 5.1, if A is saturated, then*

$$N_{R(H)}(A) = V(H) \setminus S_A.$$

Proof. No spanning path can have both endpoints in S_A: deleting S_A from such a path leaves at most $|S_A| - 1$ path segments, whereas $H - S_A$ has $|S_A|$ components. This proves that $N_{R(H)}(A) \subseteq V(H) \setminus S_A$.

Fix a Hamilton cycle C. Since $p(K) = |A|$, deleting A from C gives exactly $|A|$ nonempty paths and hence a minimum cover of K. Let $b \notin S_A$, and let Q be its path in this fixed cover. Then $b \in E(K)$, so the preceding cover-restriction result makes b eligible in the traceable cograph $H[V(Q)]$.

If Q is a singleton, delete one of its incident cycle edges. The result is a spanning path from b to a vertex of A. Otherwise let u, v be the endpoints of Q. They are a compatible endpoint pair. The complete multipartite endpoint relation in $H[V(Q)]$ makes b compatible with at least one of u, v. Replace Q by a spanning path from b to that endpoint, retain the cycle seam there, and delete the other seam from Q to A. All other portions of C remain in place. This gives a spanning H-path from b to a vertex of A, as required. □

Proposition 5.3 (Incompatible saturated profiles)**.** *Let A, B be nonempty vertex sets for which $H - A$ and $H - B$ are cographs. Suppose no spanning H-path has one endpoint in A and the other in B. If either profile is saturated, then both are saturated and $S_A = S_B$.*

Proof. Suppose A is saturated. Proposition 5.2 places B inside S_A. Since S_A is tight, Proposition 5.1 makes B saturated and gives $S_A \subseteq S_B$. Now $A \subseteq S_B$, so maximality for A gives $S_B \subseteq S_A$. The case in which B is initially saturated is symmetric. □

Corollary 5.4 (Coincidence of canonical containers)**.** *Let A, B be saturated profiles of the same Hamiltonian graph H, with $H - A$ and $H - B$ cographs. If $B \subseteq S_A$, then $S_B = S_A$. Equivalently, if $S_B \neq S_A$, the actual profile B meets $V(H) \setminus S_A = E(H - A)$.*

Proof. Both A and B lie in the tight set S_A. The path-deletion count in the proof of Proposition 5.2 excludes a spanning path with both endpoints there. Proposition 5.3 then gives $S_B = S_A$. This uses the canonical set S_A; containment of B in an arbitrary tight set does not identify that set with S_B. □

Theorem 5.5 (Protected paths under strict slack)**.** *Let C be a fixed Hamilton cycle of H, and let A be nonempty with $K = H - A$ a cograph. There is a spanning H-path with both endpoints in A if and only if $p(K) < |A|$. When the inequality is strict, the path can be chosen so that each internal vertex of A uses exactly its two incident C-edges and each endpoint uses exactly one of them. In particular, every edge of the path is an original edge of H, and no new edge incident with A is used.*

Proof. Deleting A from a spanning path with two distinct endpoints in A leaves at most $|A| - 1$ path pieces covering K. This proves necessity.

For sufficiency, induct on $|V(K)|$, allowing A to change while the vertex set and cycle C remain fixed. If C has an edge with both endpoints in A, deleting it gives the required path. This includes $K = \varnothing$. Otherwise $C - A$ consists of $q = |A|$ nonempty paths, with starts and ends given by their cyclic orientations.

If two starts are adjacent in H, delete their incoming C-seams to A and join the starts. The two resulting cycle pieces become one spanning path with endpoints in A, and every A-incidence has the required protection. The same switch works for

adjacent ends, using their outgoing seams. We may assume that the sets of starts and ends are both independent.

For each component K_j of K, let q_j count the old cycle paths it contains. These paths cover K_j, so $p(K_j) \le q_j$. Additivity of p, together with $p(K) < q$, gives a component U with $p(U) < q_j$. Hence $q_j \ge 2$ and U is a nonsingleton connected cograph. Its root decomposition is a complete join of at least two nonempty children. Independence places all old starts in U in one child X, and all old ends in one child Z.

Suppose first that $X \ne Z$. Choose distinct old paths Q_i, Q_ℓ in U, and orient Q_i from X to Z. At its first departure from X, let uv be the traversed edge, with $u \in X$ and $v \notin X$. Replace uv by the entire path Q_ℓ, traversed from its Z-end to its X-end. Both new seams are original join edges: the Z-end joins u, and the X-end joins v. Delete the two old C-seams from Q_ℓ to A. Its former location opens the cycle into one spanning path whose endpoints are the two distinct A-vertices flanking Q_ℓ. Each loses one seam and retains its other C-edge; all other A-incidences are unchanged.

We may therefore assume $X = Z$. Put $Y = V(U) \setminus V(X)$, where $Y \ne \varnothing$. All old U-path endpoints lie in X. If an old U-path has an edge within Y, inserting a different entire old U-path into that edge by the complete X-Y join, and deleting the inserted path's two old seams to A, gives the same protected spanning path. Assume that no such edge occurs.

Set $A' = A \cup Y$, and obtain H' by deleting all edges with both ends in A'. The cycle C survives: it has no A-A edge by the first case, no A-Y edge because the old U-path endpoints lie in X, and no Y-Y edge by the preceding case. Moreover, $H' - A'$ is the disjoint union of X and the other components of K, so it is a smaller cograph. Deleting Y from a minimum path cover of U gives $p(X) \le p(U) + |Y| < q_j + |Y|$. The other components require at most their old numbers of cycle paths. Thus $p(H' - A') < q + |Y| = |A'|$, and induction supplies a spanning path P protected at every vertex of A'.

It remains to move any endpoint of P in Y into the original A. Orient P from such an endpoint y. In H', a Y-vertex has no neighbor outside U: other components of K are anticomplete to U, and all A-Y edges were deleted. Every U-boundary edge of P therefore meets the original A, where protection makes it an original C-edge. The cycle has exactly $2q_j$ such boundary edges, all between A and X. An internal A-vertex retains both; an endpoint in A loses at most one. Since one endpoint is $y \in Y$, at least $2q_j - 1 \ge 3$ boundary edges survive in P.

Starting at y inside U, the boundary crossings alternate between exiting and entering U. The second crossing is an entering edge ax, with $a \in A$ and $x \in X$. The vertex a is internal to P, so its two path edges are its two C-edges. Write P as $y, \dots, a, x, \dots, z$. The join edge yx belongs to H'. Delete ax, add yx, and reverse the initial segment to obtain the single spanning path $a, \dots, y, x, \dots, z$. Its new endpoint a retains its other C-edge. The added edge is within U, and every other original A-incidence is unchanged. If $z \in A$, this finishes the proof.

Suppose instead that both endpoints of the inductive path lay in Y. Initially all $2q_j$ old U-boundary edges survived, because every original A-vertex was internal. The first rotation deleted exactly one and added none, leaving $2q_j - 1 \ge 3$ boundary edges. Reorient the modified path from its remaining Y-endpoint z toward a. Its second crossing is an entering edge a_2x_2, where $a_2 \in A$ is internal, distinct from

the terminal endpoint a, and $x_2 \in X$. Delete a_2x_2, add the original join edge zx_2, and reverse the prefix through a_2. The result is one spanning path with distinct endpoints a_2, a in A. The first keeps its other C-edge, the second keeps the edge retained after the first rotation, and every internal A-vertex keeps both C-edges. This completes the induction. □

Corollary 5.6 (One-vertex insertion)**.** *Let G be 1-tough and let $x \in V(G)$. If $H = G - x$ is Hamiltonian and $G - N_G[x]$ is a cograph, then G is Hamiltonian. In particular, a 1-tough induced-$(P_4 \cup P_1)$-free graph with a Hamiltonian vertex deletion is Hamiltonian.*

Proof. Put $A = N_G(x) \subseteq V(H)$. Toughness makes G connected, so $A \neq \varnothing$. If H had no spanning path with both endpoints in A, Theorem 5.5 and the Hamilton-cycle bound $p(H - A) \leq |A|$ would make A saturated. Proposition 5.1 would then give a tight H-set S_A containing A. Deleting S_A from the original graph G leaves the $|S_A|$ components of $H - S_A$ and the isolated vertex x, contrary to 1-toughness. A spanning A-to-A path therefore exists in H; its two endpoint edges to x close a Hamilton cycle of G.

In an induced-$(P_4 \cup P_1)$-free graph, an induced P_4 in $G - N_G[x]$ would, together with x, induce $P_4 \cup P_1$. Hence that antineighborhood is a cograph, and the first assertion applies. □

Proposition 5.7 (Toughness of a traceable extension)**.** *Let H be Hamiltonian, and let R be a nonempty vertex-disjoint traceable graph. Form the induced union J using arbitrary actual edges between H and R. Suppose every $v \in V(R)$ has a nonempty H-profile $A_v = N_H(v)$, and no tight set of H contains A_v. Then J is 1-tough.*

If the whole host J is induced-$(P_4 \cup P_1)$-free, then every $H - A_v$ is a cograph and the no-tight-container condition is equivalent to $p(H - A_v) < |A_v|$.

Proof. Write any deletion set as $S \cup T$, where $S \subseteq V(H)$ and $T \subseteq V(R)$. If $S = \varnothing$, the intact H is connected and every surviving R-vertex has an H-neighbor; the remainder is connected. If $S = V(H)$, deleting T from a fixed spanning path of R gives $c(J - (S \cup T)) = c(R - T) \leq |T| + 1 \leq |S| + |T|$.

Now suppose S is nonempty and proper. Hamiltonicity gives $c(H - S) \leq |S|$. If this inequality is strict, deleting T from a spanning path of R yields

$$c(J - (S \cup T)) \leq c(H - S) + c(R - T) \leq (|S| - 1) + (|T| + 1) = |S| + |T|.$$

If equality holds, then S is tight. Every surviving R-vertex has an H-neighbor outside S, since otherwise its profile would be contained in S. Adding these vertices creates no component beyond those of $H - S$, so $c(J - (S \cup T)) \leq |S|$. These cases prove toughness.

For the final assertion, an induced P_4 in $H - A_v$, together with the outside vertex v, would induce $P_4 \cup P_1$ in the whole host J. Thus $H - A_v$ is a cograph. Proposition 5.1, including its empty-remainder case, and $p(H - A_v) \leq |A_v|$ identify the absence of a tight container with the strict inequality. □

Theorem 5.8 (Hamiltonian prefix insertion)**.** *Let H be a Hamiltonian induced noncograph in an induced-$(P_4 \cup P_1)$-free graph W, and let R be a disjoint traceable induced subgraph. Suppose every $v \in V(R)$ has a nonempty H-profile contained in no tight H-set. Then the induced union of H and R is Hamiltonian.*

Proof. Choose a spanning path $v_1, \dots, v_r$ of R. Let $J_0 = H$, and let J_i be the subgraph of W induced by $V(H) \cup \{v_1, \dots, v_i\}$. Every nonempty prefix of the chosen path is spanning in its induced R-prefix. Its vertices retain their actual H-profiles, so Proposition 5.7 makes every J_i, $i \geq 1$, 1-tough.

Inductively assume J_{i-1} Hamiltonian, starting with the given cycle of H. The actual deletion $J_i - v_i = J_{i-1}$ is Hamiltonian. Also J_i is an induced subgraph of W. An induced P_4 in $J_i - N_{J_i}[v_i]$, including one using earlier vertices of R, would form an induced $P_4 \cup P_1$ with v_i. Thus this entire antineighborhood is a cograph. Corollary 5.6 applies to J_i and inserts v_i. Induction reaches the induced union of H and R. □

Proposition 5.9 (Strict profiles in two blocks)**.** *Let G be a 1-tough induced-$(P_4 \cup P_1)$-free graph with a spanning two-cycle 2-factor whose induced blocks are H and D. Suppose H is a noncograph and G is not Hamiltonian. Then every $v \in V(D)$ has a nonempty H-profile A_v, the graph $H - A_v$ is a cograph, and $p(H - A_v) < |A_v|$. Consequently the induced union of H with any nonempty traceable induced subgraph of D is 1-tough.*

Proof. If some $v \in D$ had no H-neighbor, an induced P_4 in the noncograph H, together with v, would induce $P_4 \cup P_1$. Thus each A_v is nonempty. The same forbidden-subgraph argument shows that A_v meets every induced P_4 of H, so $H - A_v$ is a cograph.

Suppose A_v were saturated. If v, w are consecutive on the chosen cycle of D, no spanning H-path has endpoints in A_v, A_w: such a path, the edges to v, w, and the spanning D-path obtained by deleting vw would form a Hamilton cycle of G. Proposition 5.3 therefore makes A_w saturated with $S_{A_w} = S_{A_v}$. Continuing around the D-cycle gives one tight H-set S containing every H-D attachment.

The set S is proper in H. Deleting it from the original G leaves the $|S|$ components of $H - S$ and the separate connected block D. This contradicts 1-toughness. No row is saturated, and Proposition 5.1 gives $p(H - A_v) < |A_v|$ for every $v \in D$. Every nonempty traceable induced subgraph of D inherits these same H-profiles, so Proposition 5.7 proves the final assertion. □

6. Path covers with prescribed terminal sets

The all-cograph factor argument requires a cover of a cograph whose every path ends in a prescribed independent set. We prove both existence of a minimum such cover and the interval of attainable cover sizes.

Definition 6.1 (Terminal covers)**.** Let K be a cograph and let $B \subseteq V(K)$ be nonempty and independent. A B-terminal cover is a path cover of K whose two endpoints on each path lie in B; a singleton in B is allowed. Let $z_B(K - S)$ count the components of $K - S$ disjoint from $B \setminus S$. The terminal neighbor condition is

$$b \in B, \quad v, w \in V(K) \setminus B, \quad bv \in E(K), \quad bw \notin E(K) \quad \Longrightarrow \quad vw \in E(K). \tag{4}$$

For a cover $\mathcal{P}$, its endpoint deficit is the number of its endpoints outside B, counting a singleton outside B twice.

Lemma 6.2 (Endpoint deficit with slack)**.** *Let K be a connected cograph with a nonempty independent set B satisfying (4). Let $t \geq 1$ be an integer and assume*

$$z_B(K) \leq t - 1, \qquad z_B(K - S) \leq |S| + t - 1 \quad (\varnothing \neq S \subseteq V(K)).$$

Then some minimum path cover of K has endpoint deficit at most $2t-1$.

Proof. We induct on $|K|$. If $p(K)=1$ and $t\ge 2$, any spanning path suffices. If $p(K)=t=1$, some terminal belongs to $E(K)$: otherwise $B\subseteq F(K)$, and deleting the nonempty set $F(K)$ leaves $|F(K)|+1$ terminal-free components, contrary to the assumed bound. A spanning path ending at that terminal has deficit at most one.

Suppose $p(K)\ge 2$. Choose a nontrivial complete join $K=P\vee Q$. Independence puts all of B on one side, named P. Write $r=|Q|$. Deleting Q together with any subset of P shows that P satisfies the same cut inequalities with parameter $t+r$, including the empty-cut bound $z_B(P)\le t+r-1$.

We need a minimum cover of P of deficit at most $2(t+r)-1$. If P has a terminal–nonterminal edge bv, every nonterminal w belongs to the component containing that edge: if bw is an edge use the path v,b,w, and otherwise use the edge vw from (4). All other components are isolated terminals. Restrict the cut inequalities to the mixed component and apply induction there; adding those isolated terminals changes neither deficit nor minimality. If there is no terminal–nonterminal edge, write $P=B\,\dot\cup\, W$ with no cross edges. Delete $F(W)$. The terminal-free component count is $|F(W)|+p(W)$, so the bound for P gives $p(W)\le t+r-1$, whether $F(W)$ is empty or not. A minimum cover of W, together with the singleton terminals, has deficit $2p(W)\le 2(t+r)-2$.

The join recurrence has a unique dominant side. If it is Q, deleting $P\cup F(Q)$ from K leaves $|F(Q)|+p(Q)$ terminal-free components. Thus $p(Q)\le |P|+t-1$. Chain a minimum Q-cover through the vertices of P as singleton connectors. The resulting minimum cover has $p(Q)-|P|\le t-1$ paths and deficit at most $2t-2$.

If the dominant side is P, start with the cover just constructed and insert the r vertices of Q one at a time between distinct cover paths. Each insertion lowers the number of paths by one, and there are always two paths available because $p(P)>r$. When the deficit is at least three, two distinct paths have an endpoint outside B; orient these ends toward the connector and remove both from the deficit. When the deficit is one or two, use one endpoint outside B and an endpoint of another path, reducing the deficit by at least one. When it is zero, any such merger preserves zero. Starting with deficit at most $2(t+r)-1$, these r mergers leave deficit at most $2t-1$. The path count is $p(P)-r=p(K)$, as required. □

Proposition 6.3 (Minimum terminal covers)**.** *Let K,B satisfy Definition 6.1 and* (4)*. Assume*

$$z_B(K)=0,\qquad z_B(K-S)\le |S|-1\quad(\varnothing\ne S\subseteq V(K)). \tag{5}$$

Then K has a B-terminal cover with $p(K)$ paths.

Proof. If every vertex is a terminal, use singleton paths. Otherwise (5) makes each component meet B, and some terminal–nonterminal edge exists. The argument in Lemma 6.2 places all nonterminals in one component; all remaining vertices are isolated terminals. The mixed component inherits (5). It therefore suffices to treat connected K of order at least two.

First suppose $p(K)=1$. If $B\subseteq F(K)$, the canonical endpoint cut contradicts (5). Choose a terminal $b\in E(K)$. If no spanning path has both endpoints in B, the complete multipartite relation $R(K)$ places every eligible terminal in the part P containing b, so $B\subseteq F(K)\cup P$. In the critical case of Proposition 4.8, its nonempty part cut S leaves $|S|+1$ components with all surviving terminals in one component.

At least $|S|$ components are terminal-free, contradicting (5). In the balanced case $F(K) = \varnothing$ and $B \subseteq P$; deleting P leaves $|P|$ terminal-free components, again a contradiction. Thus a B-ended spanning path exists.

Now let $p(K) \geq 2$, write $K = P \vee Q$ with $B \subseteq P$, and put $r = |Q| \geq 1$. The side Q cannot be dominant: the cut $P \cup F(Q)$ gives $p(Q) \leq |P| - 1$. Hence $p(P) > r$ and $p(K) = p(P) - r$. The cuts obtained by deleting Q give the slack bounds on P with parameter r. If P has a mixed component, apply Lemma 6.2 to it and add isolated terminals, obtaining a minimum cover of deficit at most $2r - 1$. If it has no terminal–nonterminal edge, the cut $F(P - B)$ gives $p(P - B) \leq r - 1$, and the singleton-terminal construction gives deficit at most $2r - 2$.

Insert all r vertices of Q as singleton connectors between distinct paths. Use the deficit-reducing choices in the preceding lemma. After the first $r - 1$ insertions the deficit is at most one, and the final insertion removes it; a deficit that has already reached zero stays zero. The resulting cover has $p(P) - r = p(K)$ paths, all with terminal endpoints. □

Theorem 6.4 (Terminal cover spectrum)**.** *Under the hypotheses of Proposition 6.3, a B-terminal cover of K with q paths exists precisely for the integers*

$$p(K) \leq q \leq |B| - p(K - B). \tag{6}$$

Proof. Delete B from a terminal cover. A path with b terminals leaves $b-1$ nonempty pieces, because B is independent and its two endpoints are terminals. This includes a singleton terminal, which leaves no piece. A q-path terminal cover therefore yields a cover of $K - B$ with $|B| - q$ paths, proving the upper bound; minimality gives the lower bound.

Proposition 6.3 supplies the lower endpoint. Suppose an actual terminal cover has $r \geq 1$ paths and $p(K - B) + r < |B|$. Add r fresh independent vertices, each adjacent to every vertex of B and to no other vertex. Arrange the r cover paths cyclically, with one fresh vertex between each successive pair, obtaining a Hamilton cycle of the augmented graph. When $r = 1$, the strict inequality ensures that the sole path is not a singleton, so the cycle has at least three vertices. The augmented graph minus B is the disjoint union of $K - B$ and the r fresh vertices; its path-cover number is $p(K - B) + r < |B|$.

Apply Theorem 5.5 to this cycle with protected set B. It gives a spanning path with endpoints in B. Remove the fresh vertices. They are internal, pairwise nonadjacent, and each has both neighbors in B. Their deletion leaves exactly $r + 1$ nonempty paths, all ending in B. All remaining edges are original edges of K. This raises the terminal-cover size by one. Iteration reaches every integer in (6). □

7. Factors with cograph blocks

The endpoint structure of each factor block first restricts how components outside that block can attach. The resulting separator identifies two cograph sides. The terminal spectrum then gives path covers whose endpoints can be joined through the original complete interfaces.

Theorem 7.1 (All-cograph minimum factor)**.** *Let G be a finite simple 1-tough graph of order at least three with no induced $P_4 \cup P_1$. Suppose that F is a 2-factor with the minimum number of cycles among all 2-factors of G, and that the graph induced by the vertices of each cycle of F is a cograph. Then G has a Hamilton cycle.*

Equivalently, every minimum-cycle 2-factor of a non-Hamiltonian graph under the host hypotheses has a noncograph cycle-induced block.

Proof. If F has one cycle, that cycle is Hamiltonian. Assume that it has at least two cycles, and call their induced graphs $H_1, \dots, H_s$. Each H_i is a Hamiltonian cograph of order at least three.

We first record the endpoint information needed for these blocks. Let H be any one of them, and fix its Hamilton cycle. The root join of H has at least two nonempty children, namely the induced graphs on the components of $\overline{H}$. For a proper root-join child X, put $Y = V(H) \setminus V(X)$ and $r = |Y| \geq 1$. Deleting Y from the Hamilton cycle leaves at most r nonempty path segments covering X. Hence $p(X) \leq r$; in particular, no root child is critical in the sense $p(X) > r$. The balanced case of Proposition 4.8 now says that the spanning-path endpoint graph of H is complete multipartite on $V(H)$, and that each of its independent parts P, called a *port class*, satisfies $c(H - P) = |P|$. Thus vertices in different port classes are the endpoints of a spanning path in H.

We claim that H has at most one port class of size at least two, except when $H = K_{m,m}$ for some $m \geq 2$; in that case its two sides are exactly its port classes. Let P be a port class of size at least two. Since $c(H - P) = |P| \geq 2$, the surviving graph is nonempty and disconnected. Its vertices must all lie in one root-join child X: survivors in two children would be joined by every cross edge. Set $Y = V(H) \setminus V(X)$, $r = |Y|$, and $Z = P \cap V(X)$. Every vertex of Y belongs to P, so P is the disjoint union $Y \cup Z$, while $H - P = X - Z$. Consequently

$$c(X - Z) = r + |Z|. \tag{7}$$

Choose a minimum $p(X)$-path cover of X, and delete Z from its paths. Each deletion increases the number of nonempty pieces by at most one, while deleting a singleton decreases it. The remaining pieces cover $X - Z$, so adding the other edges of $X - Z$ gives $c(X - Z) \leq p(X) + |Z|$. (7) yields $p(X) \geq r$, and the Hamilton-cycle deletion bound yields $p(X) = r$.

Suppose two distinct root children X_i, X_j have this equality. With $n = |V(H)|$ and $n_i = |V(X_i)|$, singleton path covers give $n - n_i = p(X_i) \leq n_i$, and likewise $n - n_j \leq n_j$. Thus $n_i + n_j \geq n$. The children are disjoint, so equality holds, there are no further children, and $n_i = n_j = m$. Each child has path-cover number m, hence has no edge: one edge and the remaining singleton paths would give an $(m-1)$-path cover. Therefore $H = K_{m,m}$, with $m \geq 2$ because H has a Hamilton cycle and at least three vertices. Otherwise every nonsingleton port class determines the same child X and contains its nonempty outside set Y. Distinct port classes are disjoint, so there is at most one such class.

In $K_{m,m}$, alternation forces the endpoints of every spanning path to lie on opposite sides. Conversely, for prescribed vertices a and b on opposite sides, enumerate those sides as $a = a_1, a_2, \dots, a_m$ and $b_1, \dots, b_{m-1}, b_m = b$. The word $a_1, b_1, a_2, b_2, \dots, a_m, b_m$ is a spanning path using original edges. Thus every prescribed opposite-side pair is realized, and the two sides are precisely the port classes.

We next use the minimum choice of F. Form a simple bipartite auxiliary graph with a vertex h_i for each block H_i and a distinct vertex w_e for *each physical cross edge* $e = xy$ between H_i and H_j. Its edges are $h_i w_e$ and $w_e h_j$. Color $h_i w_e$ by the pair consisting of i and the port class of x, and color $h_j w_e$ by the corresponding

pair for y. Different blocks therefore use disjoint color palettes. A properly colored auxiliary cycle selects distinct blocks and, at each selected block, endpoints in different port classes. Spanning paths with those endpoints, joined by the selected original cross edges, replace its factor cycles by one cycle. This also applies to a four-edge auxiliary cycle involving two blocks. It would reduce the number of cycles of F, so the auxiliary graph has no properly colored cycle.

We check that a cross-edge vertex cannot be the separator supplied by [Yeo97]. The graph G is connected: an empty deletion would otherwise leave at least two components and violate toughness. Every vertex has at least two neighbors, since it lies on a cycle of the given factor. If ab were a bridge, let U_a, U_b be the components of $G - ab$ containing its ends. A neighbor $u \neq b$ of a lies in $U_a - \{a\}$. Deleting a leaves u and b in different components, contrary to toughness. Thus G has no bridge.

A walk in G maps to a walk among the auxiliary block vertices by making each step inside a block stationary and replacing each physical cross edge f by its two-edge route through its own w_f. This proves that the auxiliary graph is connected. For a cross edge e, the graph $G - e$ is connected. The same walk map uses only vertices w_f with $f \neq e$; paths within its factor blocks remain available because e is a cross edge. Hence all block vertices, and then all remaining subdivision vertices, lie in one component after w_e is deleted. The two block neighbors of w_e have different colors, so w_e cannot have monochromatic edges into that component.

Yeo's separator theorem [Yeo97], which allows an unrestricted number of colors, therefore gives a block vertex h. Write H for its block and $W = G - H$. Components of W correspond to the components of the auxiliary graph minus h: routes through physical cross edges preserve connectivity in both directions, and subdivision vertices left by deleting h do not join components. All edges from h into any one such component have one color. Hence each component of W attaches to H through one port class. Here $W \neq \varnothing$, because F has at least two cycles, and every component of W has an H-neighbor, because G is connected.

No used port class is a singleton. If a component D of W attaches through $\{p\}$, deleting the original vertex p separates D from the nonempty graph $H - \{p\}$. Thus $c(G - \{p\}) \geq 2 > 1$, contrary to toughness. Nor can all components use the same larger port class P. Deleting P removes every H-W edge while leaving W intact, so the original graph satisfies

$$c(G - P) = c(H - P) + c(W) = |P| + c(W) \geq |P| + 1 > |P|. \tag{8}$$

Here $c(W) \geq 1$, and the count is at least three because $|P| \geq 2$. The classification of large port classes therefore forces $H = K_{m,m}$, with sides A, B of size $m \geq 2$. Both sides are used. If every outside component attached through A, the original cut A would leave the m independent vertices of B and the intact nonempty graph W, giving $c(G - A) = m + c(W) \geq m + 1 > |A|$. The same count with B rules out attachments solely through B.

Let U be the union of the components of W attaching through A, and V the union of those attaching through B. Both are nonempty. The sets A, B are independent and complete to each other; U is anticomplete to $B \cup V$, and V is anticomplete to $A \cup U$. Every component in U has an A-neighbor, and every component in V has a B-neighbor. These are statements about original vertices and original edges; no interface has been completed.

The graph $G[A \cup U]$ is a cograph: an induced P_4 there, together with any vertex of V, would induce $P_4 \cup P_1$. Likewise $G[B \cup V]$ is a cograph. We show that at least one of the interfaces A-U and B-V is complete. Suppose A-U is incomplete and fix $b \in B$. If some $u \in U$ has an A-neighbor a_1 and an A-nonneighbor a_0, then a_0, b, a_1, u is an induced P_4. Otherwise every U-vertex is complete or anticomplete to A. An incomplete interface supplies an anticomplete vertex, while its connected U-component has a vertex complete to A, since that component attaches to A. Along a path in the component between them, choose adjacent vertices u_1, u_0 respectively complete and anticomplete to A. For any $a \in A$, the word b, a, u_1, u_0 is an induced P_4. In either case a vertex of V is anticomplete to the other three path vertices. To avoid an induced $P_4 \cup P_1$, it must be adjacent to b. Since b was arbitrary, B-V is complete. Interchanging the sides if needed, assume henceforth that A-U is complete.

For $b \in B$, put $N_b = N_G(b) \cap V$ and $M_b = V \setminus N_b$. Every vertex of N_b is adjacent to every vertex of M_b. Indeed, if $v \in N_b$ and $w \in M_b$ were nonadjacent, then for $u \in U$ and $a \in A$, the word u, a, b, v would be an induced P_4 with w isolated from it. The sets N_b are nested. If $N_b, N_{b'}$ were incomparable, choose $v \in N_b \setminus N_{b'}$ and $w \in N_{b'} \setminus N_b$. The established completeness gives vw, and b, v, w, b' is an induced P_4 in the cograph $G[B \cup V]$. Thus the M_b are nested in reverse order. There is no edge of $\overline{G[V]}$ between N_b and M_b, so each M_b is a union of components of $\overline{G[V]}$, including when it is empty or all of V.

Put $K = G[B \cup V]$. For $S \subseteq V(K)$, the intact graph on $A \cup U$ is connected. Every component of $K - S$ meeting $B \setminus S$ joins it through the complete A–B interface, and every terminal-free component has no edge to it. Consequently

$$c(G - S) = 1 + z_B(K - S).$$

Connectedness of G gives $z_B(K) = 0$. For nonempty S, if $z_B(K - S) = 0$, then $z_B(K - S) \le |S| - 1$; otherwise the displayed count is at least two and toughness gives the same inequality. The previously proved relation between N_b and M_b is exactly (4). Thus K and B satisfy the hypotheses of Theorem 6.4.

Two further original cuts determine the cover sizes. Deleting $A \cup F(U) \cup F(K)$ leaves the disjoint remainders of U and K. Both are nonempty by Proposition 4.5, so toughness and their endpoint-cut identities give $p(U) + p(K) \le m$. Deleting $B \cup U \cup F(V)$ leaves the m independent vertices of A and the remainder of V, with no edges between them. There are at least $m \ge 2$ components, and the same cut identity gives $p(V) \le |U|$.

Set $q = \max\{p(K), m - |U|\}$ and $k = m - q$. The minimum terminal cover and the necessary bound in (6) give $p(K) + p(V) \le m$. Together with $p(V) \le |U|$, this implies $q + p(V) \le m = |B|$. The spectrum theorem gives a B-terminal cover of K with exactly q paths. Also $p(U) \le k \le |U|$: the first inequality follows from $p(U) + p(K) \le m$ and $p(U) \le |U|$, and the second from $q \ge m - |U|$. Split edges of a minimum U-cover until it has k paths.

The two covers have $k + q = m$ paths in total. Every endpoint of a U-path is adjacent to every vertex of A, as is every endpoint of a K-path, since the latter endpoints lie in B. Enumerate A and place its m vertices cyclically between the m oriented cover paths. These paths and singleton connectors are disjoint, cover all vertices, and every joining edge is present in G. Since $m \ge 2$ and U, V are nonempty, the resulting closed traversal is a simple Hamilton cycle. □

8. Protected restoration and cycle assemblies

This section uses the saturated-profile and cograph path-cover results established earlier in the host paper. The strict and equal-count cases require different endpoints: strict saving releases two vertices of the original cycle, while equal count releases an ineligible vertex of its cograph remainder.

Setup 8.1. Throughout this section, unions retain their actual edges. The symbols p, E, F, D, R have the meanings in Definition 4.1. We use the endpoint and restriction results of Section 4, and the saturated profiles and insertion rule of Section 5. A path is protected at A relative to a specified cycle if every path edge incident with A is an edge of that cycle.

Theorem 8.2 (Strict protected absorption)**.** *Let H have a fixed Hamilton cycle C, let $A \neq \varnothing$, and suppose $K_0 = H - A$ is a cograph with $p(K_0) = |A| = a$. Let D be vertex-disjoint from H and have a fixed Hamilton cycle. Assume that $K = K_0 + D$ is a cograph and every vertex of D has a neighbor in $E(K_0)$. If $p(K) < a$, then $H + D$ has a spanning path with both endpoints in A. Every internal vertex of A retains both its incident C-edges, each endpoint retains one incident C-edge, and every edge of the path is original.*

Proof. The saturated-profile conclusion in Setup 8.1 excludes a spanning A-to-A path in H. Hence C has no A-A edge, and $C - A$ is a minimum a-path cover of K_0. Its starts are independent, as are its ends: an edge between two endpoints with the same cyclic sign would join two pieces of C into the excluded spanning path.

The connected graph D lies in one component U of K. Its eligible old neighbors ensure that U contains old vertices. Write $U_0 = U \cap K_0$; it is a union of whole components of K_0. If q old paths lie in U_0, minimum-cover restriction gives $p(U_0) = q$. Every other component is unchanged, so $p(U) < q$ and $q \geq 2$.

Consider the root join of the connected cograph U. All old starts belong to one child X, since starts in different children would be adjacent. All old ends similarly belong to one child Z. Suppose $X \neq Z$. Orient an old path from X to Z, and take its first edge leaving X. A different whole old path can be inserted into that edge, using its Z-end and X-end as the two join ports. Both new seams join old vertices and are therefore edges of H. Removing the inserted path's two old A-seams yields a spanning A-to-A path in H, a contradiction. Thus $X = Z$; put $Y = U - X$. The same insertion into an old Y-Y edge shows that no such edge exists. Every old vertex of Y is consequently an internal singleton connector on an old path.

Put $X_0 = X \cap K_0$ and $Y_0 = Y \cap K_0$. If $Y_0 \neq \varnothing$, then $U_0 = X_0 \vee Y_0$ is one component of K_0. The q old paths have both ends in X_0 and use every vertex of Y_0 as a singleton connector, so $|X_0| \geq q + |Y_0|$. The join recurrence, $p(U_0) = q \geq 2$, and $p(Y_0) \leq |Y_0|$ give

$$p(X_0) = q + |Y_0|.$$

The saturated-join endpoint description puts Y_0 in $F(K_0)$ and inherits eligibility in U_0 from X_0. These identities also hold when $Y_0 = \varnothing$ as well. Thus every eligible old neighbor of a vertex of D in U lies in X.

Suppose first that D meets Y. Break a fixed D-cycle edge de with $d \in Y$, choosing a crossing edge if D also meets X. This gives a spanning D-path from e to d. If $e \in Y$, join its ends to the starts of two different old U-paths. The joins are actual X-Y edges. Deleting the corresponding incoming A-seams leaves one spanning A-to-A path, while every other A-incidence stays on C.

If $e \in X$, choose an eligible old neighbor $u \in X$ of e. Let Q_i be its old path. Minimum-cover restriction makes u locally eligible on Q_i. The traceable-cograph endpoint relation supplies a spanning Q_i-path from u to one of its old endpoints; for a singleton Q_i, either cyclic seam can be released. Keep that endpoint's old A-seam and expose u at the other cyclic position. On a different old path Q_j, choose the endpoint with the same cyclic sign. It lies in X and is adjacent to d. The original edges ue and dQ_j join the exposed pieces through the spanning D-path. Delete the matching old A-seams at Q_i and Q_j. The equal signs make the result one path, with two distinct original A-vertices as endpoints and the asserted protection.

It remains to treat $D \subseteq X$. Every vertex of Y is then old. The saturated-join calculation gives $p(K_0 - Y) = a + |Y|$ and $E(K_0 - Y) = E(K_0)$. Set $A' = A \cup Y$, and delete every edge internal to A' from $H + D$. The original H-cycle survives: it has no A-A, A-Y, or Y-Y edge. The D-cycle survives as well. The old remainder $K_0 - Y$ is saturated at A', and every vertex of D still sees its eligible set. At the root join,

$$p(X) \le p(U) + |Y| < q + |Y|,$$

so the unchanged components give $p(K - Y) < a + |Y| = |A'|$. Induction on $|V(K)|$ supplies a spanning path protected at A', with both endpoints in A'. This induction uses only the stated graph and cograph hypotheses; the edge-deleted intermediate graph need not remain induced-$(P_4 \cup P_1)$-free.

If both endpoints lie in A, the path is ready. Otherwise orient it from an endpoint $y \in Y$. The original cycle has exactly $2q \ge 4$ crossings between U and its complement, all old A-X edges. In the edge-deleted graph, Y has no edge outside U, and distinct components of K have no edge between them. Every used crossing therefore has an original A-end and, by protection, is an old C-edge. At most one original A-vertex is a path endpoint, so at least $2q - 1 \ge 3$ crossings survive. Starting at y, the second crossing is an entering edge ax, with $a \in A$ internal and $x \in X$. In the path word $y, \dots, a, x, \dots, z$, replace ax by the original join edge yx and reverse the prefix. The new word is $a, \dots, y, x, \dots, z$. Vertex a keeps its other C-edge, and no other A-incidence changes.

If $z \in A$, this finishes. If $z \in Y$, all original A-vertices were internal before the first rotation, so all $2q$ crossings initially survived and at least $2q - 1 \ge 3$ remain. Reorient from z toward a. A second entering edge a_2x_2 exists; a_2 is an internal original A-vertex distinct from the terminal a. Replace it by the original join edge zx_2 and reverse that prefix. The endpoints are now the distinct vertices a_2, a of A, each retaining one C-edge. Every internal original A-vertex retains both. The second rotation requires protection only at original A, which the first rotation preserved. □

Theorem 8.3 (Rooted-prefix restoration)**.** *Let H, D, X be pairwise vertex-disjoint induced blocks, each with an original Hamilton cycle, in one finite induced-$(P_4 \cup P_1)$-free host. Assume H is noncograph. For each $d \in D$, assume its nonempty row $C_d = N_H(d)$ has cograph complement, is saturated, and has vertexwise largest saturated container $S_{C_d} = T$. For each $y \in X$, assume its row $A_y = N_H(y)$ is saturated with $S_{A_y} = S \neq T$. If $H + D + x$ is Hamiltonian for one $x \in X$, then $H + D + X$ is Hamiltonian. Edges between X and D are permitted.*

Proof. Fix $d \in D$. Every X-row meets $H - T$. For if $A_y \subseteq T$, then A_y and C_d lie in the same tight H-set. The earlier foreign-row path obstruction and its canonical-gate

consequence in Setup 8.1 give $S_{A_y} = S_{C_d} = T$, contrary to $S \neq T$. The rows of both groups are nonempty.

Put $Q = H + D$, and break an incident edge of the fixed X-cycle to obtain a spanning X-path starting at x. We show that $Q + R$ is 1-tough for every induced prefix R of this path. Write a deletion set as $U \cup W$, where $U \subseteq Q$ and $W \subseteq R$. If $x \notin W$ and $U \neq \varnothing$, deletion of U from the Hamilton cycle of $Q + x$ leaves at most $|U|$ pieces. Deletion of W from the prefix path leaves at most $|W| + 1$ pieces, one containing x; the others add at most $|W|$ components. If $U = \varnothing$, intact H and D form a connected base, and every surviving prefix vertex meets H, so the remainder is connected.

Suppose $x \in W$. Because x was an endpoint of the prefix path, $R - W$ has at most $|W|$ pieces. If U meets both H and D, deletion from their two cycles leaves at most $|U|$ base components. If U misses H, intact H connects every surviving D- and R-vertex. The only remaining case has $\varnothing \neq U \subseteq H$ and D intact. Its base has at most $|U| + 1$ components. When it has at most $|U|$, the prefix-piece bound suffices. Equality $|U| + 1$ requires U to be tight in H and to contain every H-D attachment. A fixed D-row then lies in U, so the maximality of its canonical gate gives $U \subseteq T$. Every surviving prefix vertex has an $H - T$ neighbor, hence meets $H - U$ and adds no component. Since $x \in W$, $|U| + 1 \leq |U| + |W|$. Full deletion of H belongs to the preceding nonexceptional count. This proves 1-toughness for every prefix.

Start with the assumed Hamiltonian graph $Q + x$, and add the remaining prefix vertices one at a time. At each addition, deletion of the new vertex is exactly the preceding Hamiltonian induced prefix. The new graph remains induced-$(P_4 \cup P_1)$-free. Its whole antineighborhood at the new vertex is a cograph, since an induced P_4 there together with that vertex would be a forbidden $P_4 \cup P_1$. The toughness just proved and the earlier one-vertex insertion result in Setup 8.1 therefore give a Hamilton cycle after each addition. The final prefix is $H + D + X$. □

Corollary 8.4 (Strict saving merges three blocks)**.** *In the setting of Theorem 8.3, choose $x \in X$, put $A = N_H(x)$, and assume additionally that x is anticomplete to D. Then $K_0 = H - A$ is a cograph with $p(K_0) = |A|$. Every vertex of D sees $E(K_0) = H - S$, and $K = K_0 + D$ is a cograph. If $p(K) < |A|$, then $H + D + X$ is Hamiltonian. Other vertices of X may have neighbors in D.*

Proof. Saturation of A gives the assertion about K_0 and identifies $E(K_0) = H - S$. Fix a saturated D-row C_d. If it missed $H - S$, then $C_d \subseteq S$. The foreign-row obstruction and canonical-gate consequence of Setup 8.1, applied to A and C_d, would give $S_{C_d} = S_A$, contrary to $T \neq S$. Thus every $d \in D$ has an eligible old neighbor. The explicit anticompleteness assumption puts all of $K_0 + D$ in the antineighborhood of x; induced-$(P_4 \cup P_1)$-freeness makes K a cograph. Theorem 8.2 supplies a spanning A-to-A path of $H + D$. Its distinct endpoints both meet x, so closing through x gives a Hamilton cycle of $H + D + x$. Theorem 8.3 restores all of X. Distinct gate names alone were not used to infer the chosen vertex's anticompleteness to D. □

Lemma 8.5 (Distinct-cover released endpoints)**.** *Let L, J be cographs with $V(J) = V(L) \dot\cup V(D)$, where D induces a Hamiltonian graph, and suppose $p(L) = p(J) = q$. Let $h \in F(L) \cap E(J)$. If u, v are endpoints of distinct paths of one minimum L-cover,*

then h is $D(J)$-compatible with at least one of u, v. No condition on edges between D and L is required.

Proof. Induct on $|J|$. There is no distinct-path pair when $q = 1$. If J is disconnected, D lies in one component U, while the other components are unchanged. Additivity gives $p(L \cap U) = p(U)$, and a released h lies in U. An old endpoint outside U remains eligible and is compatible with h on distinct paths. If both u, v lie in U, restrict their minimum cover to U and apply induction. In particular, $p(U) = 1$ cannot contain both endpoints on distinct old paths.

Suppose J is connected. As $q \geq 2$, its root join has a unique critical child X. Put $Y = J - X$ and $r = |Y|$; the join recurrence gives $p(X) = q + r$. Use subscripts to denote intersection with X and Y. If $D_Y \neq \varnothing$, delete L_Y from a minimum L-cover and D_Y from a Hamilton cycle of D. Restriction and the join recurrence give

$$q + r = p(X) \leq p(L_X) + p(D_X) \leq q + |L_Y| + |D_Y| = q + r.$$

Thus equality holds throughout. A minimum L_X-cover together with a minimum D_X-cover is a minimum X-cover. The restriction theorem yields $E(X) \cap L_X = E(L_X)$, while the saturated join $L_X \vee L_Y$ gives $E(L) = E(L_X)$, also when $L_Y = \varnothing$. Since $E(J) = E(X)$, no old vertex is released, a contradiction.

Hence $D \subseteq X$ and $Y \subseteq L$. As $p(D) = 1$, $p(L_X) \geq p(X) - 1 = q + r - 1 > r$. No other root child can be critical in L, and its join recurrence gives $p(L_X) = q + r$. Eligibility and distinct-path compatibility in L and J are inherited from L_X and X, respectively. Their inherited h, u, v satisfy the induction hypotheses inside X, which proves the assertion. □

Lemma 8.6 (Traceable released endpoints)**.** *Under the partition assumptions of Lemma 8.5, suppose both L and J are traceable and $h \in F(L) \cap E(J)$. For every spanning L-path with endpoints u, v, J has a spanning h-to-u path or a spanning h-to-v path. In particular, $|V(L)| \geq 3$.*

Proof. The order bound follows from $F(L) \neq \varnothing$. First suppose J has a critical root child X, with $Y = J - X$, $|Y| = r$, and $p(X) = r + 1$. If D met Y, the equality calculation in the proof of Lemma 8.5, now with $q = 1$, would exclude a released vertex. Thus $D \subseteq X$. The join recurrence gives $r \leq p(L_X) \leq r + 1$. At the lower value no root child is critical in L, so $F(L) = \varnothing$, again impossible. Hence $p(L_X) = r + 1$. The spanning-path endpoint relations $R(L)$ and $R(J)$ are inherited from $D(L_X)$ and $D(X)$. The old endpoints u, v are $D(L_X)$-compatible. Lemma 8.5, applied inside X, makes h compatible with one of them, giving the required spanning path in J.

Now suppose J is balanced. It is Hamiltonian and $E(J) = V(J)$. If h were incompatible with both u and v as spanning-path endpoints, the complete multipartite endpoint relation would put all three in one port class P, with $c(J - P) = |P|$. Delete P from the given spanning L-path and from a Hamilton cycle of D. If $P \cap D \neq \varnothing$, the resulting forests have at most $|P \cap L| - 1$ and $|P \cap D|$ components, contradicting that equality. Thus $P \subseteq L$. Equality now forces $c(L - P) = |P| - 1$, and D is a whole component of $J - P$, with no edge to $L - P$.

We show that deleting this whole component makes every vertex of P eligible in L. Since $|P| \geq 2$, $J - P$ is disconnected. Let X be the root join child containing it, and write $Y = J - X$, $r = |Y|$, and $P = Y \cup Z$. The tight-cut identity and Hamiltonicity of J give $p(X) = r$ and $Z \subseteq F(X)$. The set $Y \cup F(X)$ is itself tight

and consists of mutually incompatible Hamilton endpoints. It contains the full port class P, so $P = Y \cup F(X)$. Put $f = |F(X)|$. Then $X - F(X)$ has $r + f$ components, one of which is precisely D.

Choose a minimum X-cover with an endpoint in D, possible because $D \cap F(X) = \varnothing$. Deleting $F(X)$ from a minimum cover leaves the components of $X - F(X)$ as individual path segments. Thus D is the whole initial segment of the selected cover path, and removing it leaves at most r paths: $p(X - D) \le r$. Counting $F(X)$ and one vertex from each remaining component also gives $|X - D| \ge r + 2f - 1$.

If $f \ge 1$, then $|X - D| \ge r + 1$. In $L = (X - D) \vee Y$, the first child has path-cover number at most r, while every Y-child has order at most $r < |X - D|$. No child is critical. Hence L is balanced and $P \subseteq E(L)$. If $f = 0$, then $P = Y$ and $|X - D| \ge r - 1$. A critical child $Z' \subseteq Y$ would require

$$|Z'| \ge p(Z') = |X - D| + (r - |Z'|) + 1 \ge 2r - |Z'|.$$

Equality forces $Z' = Y$, $|X - D| = r - 1$, and $p(Y) = r$. The $r - 1$ remaining components of $X - D$ then have $r - 1$ vertices, so the sole critical case is the complete bipartite graph $K_{r,r-1}$; its ineligible vertices lie in $X - D$, away from P. In every other case L is balanced. Again $P \subseteq E(L)$, contradicting $h \in F(L)$. Therefore h is compatible with u or v, as required. □

Theorem 8.7 (Protected released-endpoint absorption)**.** *Let H have a fixed Hamilton cycle C, let $A \neq \varnothing$, and let $K_0 = H - A$ be a cograph with $p(K_0) = |A|$. Add a disjoint Hamiltonian induced graph D. Assume $J = K_0 + D$ is a cograph and $p(J) = p(K_0)$. If $h \in F(K_0) \cap E(J)$, then $H + D$ has a spanning A-to-h path. Every internal A-vertex retains both incident C-edges, its one A endpoint retains one incident C-edge, and no new A-incidence is used.*

Proof. Induct on $|J|$. Saturation makes $C - A$ a minimum K_0-cover. The Hamiltonian graph D lies in one component U of J, with every other component unchanged. Since h is released, $h \in U$. Put $L = K_0 \cap U$. Additivity gives $p(L) = p(U) = q$, and exactly q old $C - A$ paths lie in L.

If $q = 1$, the old path spans L. Lemma 8.6 gives a spanning U-path from h to one of its old endpoints. Retain that endpoint's old A-seam and delete the other seam. Together with the untouched part of C, this is the required spanning A-to-h path.

Suppose $q \ge 2$, and write the critical root join of U as $X \vee Y$. The equality argument in Lemma 8.5 shows that $D \subseteq X$ and $Y \subseteq L$, with $p(L \cap X) = p(X) = q + |Y|$. Every old path has its endpoints in $L \cap X$, and every Y-vertex is an internal singleton connector. Set $A' = A \cup Y$ and delete edges internal to A'. The original cycle survives, since it has no A-A, A-Y, or Y-Y edge. The smaller old remainder $K_0 - Y$ is saturated at $|A'|$; the new remainder $J - Y$ has that same path-cover number. Critical-join eligibility preserves $h \in F(K_0 - Y) \cap E(J - Y)$. Induction gives a spanning A'-to-h path protected at A'.

Only an endpoint $y \in Y$ needs correction. Orient the path from y. The original cycle has $2q \ge 4$ crossings between U and its complement, all old A-to-$(L \cap X)$ edges. Neither current endpoint is in original A, and protection retains every crossing. No extra edge at A' was used. Thus an entering crossing ax occurs after an exit, with $a \in A$ and $x \in L \cap X$. Replace ax in the word $y, \dots, a, x, \dots, h$ by the original join edge yx, reversing the prefix. The resulting word $a, \dots, y, x, \dots, h$ is one spanning path. Its new A endpoint keeps the other old C-edge, and all remaining A-incidences are unchanged. □

Corollary 8.8 (Three-cycle merger from an adjacent released row)**.** *Let H, D, Q be pairwise vertex-disjoint induced blocks with specified Hamilton cycles. Let x, z be consecutive on the specified cycle of Q, and set $A = N_H(x)$. Assume $H - A$ and $(H - A) + D$ are cographs and*

$$p(H - A) = p((H - A) + D) = |A|.$$

If z is adjacent to some $h \in F(H - A) \cap E((H - A) + D)$, then $H + D + Q$ is Hamiltonian. Each specified cycle has order at least three, as in our standing convention.

Proof. The equality $p((H - A) + D) = |A|$ and nonemptiness of D give $|A| \geq 1$. Theorem 8.7 gives one spanning path P of $H + D$, from some $a \in A$ to h. Break the original Q-cycle edge xz, obtaining its spanning path from x to z. The edges ax and zh are original by the definitions of A and h. Traversing the Q-path and then the same physical path P in reverse gives the Hamilton cycle word

$$a,\ x,\ [\text{interior of the } Q\text{-path}],\ z,\ h,\ [\text{interior of } P \text{ in reverse}],\ a.$$

The construction uses one chosen A-to-h path throughout; no endpoints from different covers are combined. □

9. Locality of foreign-cycle changes

The following theorem detects a saving or release caused by a family of added cycles in one single-cycle extension. Edges between added cycles are permitted. All neighbors in its hypotheses belong to the eligible set of the original graph L.

For complete joins we use (1), with saturated endpoint inheritance from Proposition 4.4. Neither toughness nor exclusion of $P_4 \cup P_1$ is needed here.

Theorem 9.1 (Locality of savings and endpoint release)**.** *Let $L, D_1, \ldots, D_k$, where $k \geq 1$ and $L \neq \varnothing$, be pairwise vertex-disjoint induced blocks. Equip each D_i with a specified Hamilton cycle, and allow edges between distinct D_i. Let J be their induced union, and assume that J is a cograph. Assume that every vertex of every D_i has a neighbor in the original set $E(L)$.*

Then:

1. *If $p(J) < p(L)$, then $p(L + D_i) < p(L)$ for some i.*
2. *If $h \in F(L) \cap E(J)$, then, for some i, either $p(L + D_i) < p(L)$ or $h \in E(L + D_i)$.*

In the second alternative, if no single-cycle extension gives a strict saving, that extension has path-cover number $p(L)$.

Proof. For every i, break an edge of the specified cycle on D_i and append the resulting spanning path at an actual neighbor in $E(L)$, using a minimum cover that witnesses that old endpoint. This gives $p(L+D_i) \leq p(L)$. The construction concerns one added cycle at a time and uses no eligibility assertion after an extension.

We first record how to restrict the problem to a component. Every connected component of L, and every D_i, lies wholly in a component of J. A component containing a new vertex also contains an old vertex, since that vertex has an $E(L)$-neighbor. Minimum covers of a disjoint union are assembled from minimum covers of its components: paths cannot cross between components, and replacing a nonminimum constituent would shorten the whole cover. Thus path counts add and eligibility is componentwise; in particular, $E(L \cap U) = E(L) \cap U$ for a component

U of J. After adding one D_i in U, the old components outside U remain untouched and anticomplete to the part in U. A saving or endpoint release in U therefore transfers to the corresponding single-cycle extension of L. A component without a new cycle is unchanged and cannot supply either event.

Here is the old-side alignment needed when J is connected and $q = p(L) \geq 2$. Choose one minimum q-cover of L. Endpoints of distinct cover paths cannot be adjacent, since their edge would join those paths. Choose a root child X of J containing an endpoint of the first path, and put $Y = J - X$, $X_0 = L \cap X$, $Y_0 = L \cap Y$, and $r_0 = |Y_0|$. The root children of a connected cograph are its nonempty complement-components and are pairwise complete. Both endpoints of every other cover path must therefore lie in X. An endpoint of a second path similarly forces the first path's other endpoint into X. Hence every cover endpoint lies in X.

No edge with both ends in Y_0 is used by this cover. If a path used yy' there, insert a different entire cover path $Q = (q_1, \ldots, q_\ell)$ in place of that edge, in the order $y, q_1, \ldots, q_\ell, y'$. Both endpoints of Q lie in X, so the two seams are original join edges. This merges two paths, even when $\ell = 1$, when the inserted vertex occurs once. Every Y_0-vertex is consequently an internal singleton connector. Deleting them leaves exactly $q + r_0$ nonempty X_0-path pieces, and $|X_0| \geq q + r_0$.

If $Y_0 \neq \varnothing$, then $L = X_0 \vee Y_0$ literally, while $p(Y_0) - |X_0| \leq r_0 - (q + r_0) = -q < 0$. Since $p(L) = q \geq 2$, (1) forces $p(X_0) = q + r_0$. This equality permits saturated-join inheritance, giving $E(L) = E(X_0)$ and $F(L) = Y_0 \cup F(X_0)$. When $Y_0 = \varnothing$, the same formulas hold with $X_0 = L$ and $r_0 = 0$. In either case, all old eligible vertices lie in X.

A specified cycle D_i meeting Y now gives an actual strict saving. If it lies wholly in Y, breaking one cycle edge gives a spanning path whose two ends join endpoints in X of two distinct paths of the chosen old cover. If it meets both sides, choose an edge dz of its specified cycle with $d \in X$ and $z \in Y$, and break that edge to obtain a spanning d-z path. The assumed old neighbor u of d lies in $E(X_0)$. Choose one entire minimum cover of L witnessing u as an endpoint, and choose an endpoint w of a different path of this same cover. Every endpoint of this cover lies in X. The actual attachment ud, the spanning path from d to z, and the original join edge zw merge the two old paths. All three vertex blocks are disjoint, so the unchanged old paths complete a $(q-1)$-cover of $L + D_i$. The cover witnessing u need not be the cover used to establish the alignment.

We prove the strict-saving assertion by strong induction on $|J|$. If J is disconnected, additivity yields a proper component U with $p(U) < p(L \cap U)$. The component restriction gives it a nonempty old part, at least one whole new cycle, and the original-neighbor condition. Induction in U, followed by adjoining the untouched old components, proves the assertion. Thus suppose that J is connected. As $p(J) \geq 1$, a strict saving requires $q = p(L) \geq 2$. Apply the alignment above. A cycle meeting Y gives the desired witness. Otherwise every new cycle lies in X, so every vertex of the nonempty outside side Y is old. Put $r = |Y|$. We have $p(X_0) = q + r$, $E(X_0) = E(L)$, and $|X_0| \geq q + r$. Applying (1) to $J = X \vee Y$ gives $p(X) \leq p(J) + r < q + r$. The smaller graph X, with old part X_0, contains all new cycles and their eligible old neighbors. Induction gives $p(X_0 + D_i) < p(X_0) = q + r$ for some i.

Set $Z = X_0 + D_i$. The single extension is exactly $L + D_i = Z \vee Y$. Since $p(Z) \le q + r - 1$, $p(Y) \le r$, and $|Z| \ge |X_0| \ge q + r$, its other-side excess satisfies $p(Y) - |Z| \le -q < 0$. Formula (1) gives $p(L + D_i) \le q - 1$; here $q \ge 2$ also handles its constant term. Every recursive graph is smaller, and a component with no new cycle never creates an induction obligation.

For endpoint release, we need one balanced enlargement when the old count is one. Suppose a root child X and its outside Y satisfy $L_X = L \cap X \ne \varnothing$, $L_Y = L \cap Y$, $r = |L_Y|$, $p(L_X) = r + 1$, $E(L) = E(L_X)$, and $|L_X| \ge r + 1$. Let a specified cycle D meet Y, and write $D_X = D \cap X$, $D_Y = D \cap Y$. The single extension is the literal binary join $P \vee Q$, where $P = L_X + D_X$ and $Q = L_Y + D_Y$; both sides are nonempty.

If $D_X \ne \varnothing$, deleting the nonempty D_Y from the original cycle leaves a path cover of D_X with at most $|D_Y|$ paths. An endpoint d of a minimum D_X-cover has an assumed neighbor in $E(L) = E(L_X)$. Joining the corresponding paths of independent minimum covers gives $p(P) \le p(L_X) + p(D_X) - 1 \le r + |D_Y| = |Q|$. If $D_X = \varnothing$, the same bound follows from $p(P) = r + 1 \le r + |D_Y|$. For the other side, ignoring possible old-new edges gives $p(Q) \le p(L_Y) + p(D_Y)$. When $D_X \ne \varnothing$, deletion from the cycle gives $p(D_Y) \le |D_X|$, so $p(Q) \le r + |D_X| < |L_X| + |D_X| = |P|$. When $D_X = \varnothing$, $D_Y = D$ is Hamiltonian, and $p(Q) \le r + 1 \le |L_X| = |P|$. Both balanced inequalities hold. The balanced-join rule gives a Hamilton cycle of $L + D$, which has at least four vertices. Deleting a cycle edge incident with any prescribed vertex makes that vertex an endpoint of a one-path minimum cover.

We now prove the endpoint assertion by strong induction on $|J|$. Suppose, contrary to it, that every single extension has path-cover number $p(L)$, none makes h eligible, and $h \in F(L) \cap E(J)$. The strict-saving assertion just proved gives $p(J) \ge p(L)$. If J is disconnected, its component U containing h must contain new vertices; otherwise componentwise eligibility could not change. The component restriction transfers the equal-count and non-release conditions to the proper component U, where induction is a contradiction. Hence J is connected.

First suppose $q = p(L) \ge 2$. The alignment applies, and no new cycle can meet Y. Thus every new cycle lies in X, Y is entirely old, and $r = |Y| \ge 1$. We have $p(X_0) = q + r$, $|X_0| \ge q + r$, $E(L) = E(X_0)$, and $F(L) = Y \cup F(X_0)$. Since $|X| \ge |X_0| > r$ and $p(Y) \le r$, the Y-side excess in (1) is negative. The inequality $p(J) \ge q \ge 2$ forces $p(X) = p(J) + r$. Saturated-join inheritance, now justified by this numerical equality, gives $E(J) = E(X)$ and $F(J) = Y \cup F(X)$. Consequently $h \in X \cap F(X_0)$.

For each $Z_i = X_0 + D_i$, the join $L + D_i = Z_i \vee Y$ has $|Z_i| \ge |X_0| > r$ and negative Y-side excess. Its known path-cover number is $q \ge 2$, so (1) gives $p(Z_i) = q + r = p(X_0)$. Saturated-join inheritance for this single extension gives $E(L + D_i) = E(Z_i)$. Thus h stays ineligible in every Z_i. All new vertices retain neighbors in $E(X_0) = E(L)$, while $h \in F(X_0) \cap E(X)$. These are the release-induction hypotheses in the strictly smaller graph X, a contradiction.

It remains to consider $p(L) = 1$. Since $h \in F(L)$, the old graph has at least three vertices: in a graph of order one or two with a spanning path, every vertex is an endpoint. Suppose first that L meets at least two root children J_i of J, and discard empty intersections $L_i = L \cap J_i$. Their cross edges are complete, so they form an explicit join decomposition of L. If $p(L_i) \le |L - L_i|$ for every i, the balanced-join rule would make L Hamiltonian and every old vertex eligible. Hence some L_i satisfies

$p(L_i) > |L - L_i|$. The join recurrence with $p(L) = 1$ forces $p(L_i) = |L - L_i| + 1$. Set $X = J_i$, $L_X = L_i$, $L_Y = L - X$, and $r = |L_Y| \geq 1$. Only after this equality is established do we use saturated-join inheritance to obtain $E(L) = E(L_X)$ and $F(L) = L_Y \cup F(L_X)$; also $|L_X| \geq p(L_X) = r + 1$.

If instead all of L lies in one root child X, take $L_X = L$, $L_Y = \varnothing$, and $r = 0$. The same equalities and size bound hold directly. In either old-position case, the balanced enlargement above shows that any specified cycle meeting $Y = J - X$ would release h, contrary to the assumption. All new cycles therefore lie in X, and every vertex of Y is old. As $Y \neq \varnothing$, the case $r = 0$ is impossible. Hence $Y = L_Y$ and $r \geq 1$.

For each $Z_i = L_X + D_i$, appending a spanning D_i-path at an $E(L_X)$-neighbor gives $p(Z_i) \leq r + 1$. If $p(Z_i) \leq r = |Y|$, then $p(Y) \leq r < |L_X| \leq |Z_i|$. These are both balanced inequalities for $L + D_i = Z_i \vee Y$, so its Hamilton cycle would release h. Therefore $p(Z_i) = r + 1$ for every i. Applying the strict-saving assertion inside X gives $p(X) \geq p(L_X) = r + 1$, since a strict saving in X would contradict those single-extension equalities.

Now $|X| \geq |L_X| \geq r + 1$ and $p(Y) \leq r$, so the Y-side excess of $J = X \vee Y$ is negative. Formula (1) yields $p(J) = p(X) - r$. Saturated-join inheritance gives $E(J) = E(X)$ and $F(J) = Y \cup F(X)$. Thus h lies in X, and the old inheritance formula puts it in $F(L_X)$. Likewise, $p(Z_i) = r + 1$ and $p(L + D_i) = 1$ justify saturated-join inheritance for each $Z_i \vee Y$, giving $E(L + D_i) = E(Z_i)$. The assumed non-release passes to every Z_i. All new vertices still see $E(L_X) = E(L)$. Release induction applies in the strictly smaller graph X, whose old path-cover number is $r + 1 \geq 2$, and gives the final contradiction. □

10. Two cycles and noncograph unions

We establish two unrestricted statements before fixing a non-Hamiltonian host or a minimum-cycle factor in that host.

Theorem 10.1 (Two-cycle factor)**.** *Let G be a finite simple 1-tough graph of order at least three with no induced $P_4 \cup P_1$. If G is given a spanning 2-factor consisting of two cycles, then G has a Hamilton cycle made of original edges.*

Proof. Suppose that G is not Hamiltonian. The given two-cycle factor then has the minimum possible number of cycles. If both induced blocks are cographs, Theorem 7.1 gives a Hamilton cycle. Otherwise let H be a noncograph block and D the other block.

Proposition 5.9 makes every D-row nonempty and strict. By Proposition 5.1, none lies in a tight H-set. Break an edge of the specified D-cycle to obtain a spanning path and apply Theorem 5.8. The resulting Hamilton cycle of $H + D = G$ is a contradiction. □

The next lemma needs no toughness hypothesis on its ambient graph.

Lemma 10.2 (Toughness of noncograph unions)**.** *Let $H_1, \ldots, H_r$ be nonempty pairwise vertex-disjoint induced subgraphs of a graph with no induced $P_4 \cup P_1$. Suppose that each H_i has a Hamilton cycle and contains an induced P_4. Then the subgraph J induced by their union is 1-tough.*

Proof. Fix a Hamilton cycle in each H_i, and let $X \subseteq V(J)$ leave at least two components. Every component of $J - X$ is a cograph. An induced P_4 in one

component, together with a vertex of another, would give the forbidden induced graph.

The set X meets every H_i. If some block were intact, its connected vertex set would lie in one component of $J - X$ containing its induced P_4. Put $s_i = |X \cap V(H_i)| > 0$. Deleting those s_i vertices from the fixed Hamilton cycle of H_i leaves at most s_i nonempty paths, and leaves none if the entire block is deleted. These paths cover $J - X$ with at most

$$\sum_i s_i = |X|$$

paths. Adding the other edges of $J - X$ cannot increase the number of components, so $c(J - X) \le |X|$. The argument also excludes an empty separating cut, because an intact noncograph block would lie in one of its components. □

11. The minimum-factor normal form

The two unrestricted results determine the possible blocks of a non-Hamiltonian minimum-cycle factor.

Proposition 11.1 (Canonical groups)**.** *Let G be a finite simple non-Hamiltonian 1-tough graph of order at least three with no induced $P_4 \cup P_1$. Let $\mathcal{F}$ be a given 2-factor having the minimum number of cycles among all 2-factors of G, and assume it has at least two cycles. Its blocks are the subgraphs induced by the vertex sets of its specified factor cycles.*

Then $\mathcal{F}$ has at least three cycles and exactly one noncograph block H. For every other block D and every $x \in V(D)$, the row $A_x = N_H(x)$ is nonempty, $H - A_x$ is a cograph, and

$$p(H - A_x) = |A_x|, \qquad S_{A_x} = S_D.$$

Here S_D is one nonempty proper tight H-set and the unique largest tight container of each row of D. Consecutive vertices on the specified factor cycle of D have incompatible H-profiles.

Satellites with distinct gates are anticomplete. Grouping the satellites by their canonical gates gives at least two groups, each of which is a cograph.

Proof. Theorem 10.1 excludes a factor with two cycles, so $\mathcal{F}$ has at least three. Theorem 7.1 excludes the case in which every block is a cograph.

Suppose two blocks H_1, H_2 are noncographs. Lemma 10.2 makes their induced union 1-tough. That union inherits the absence of an induced $P_4 \cup P_1$ and has the given two-cycle spanning factor formed by the specified cycles of H_1, H_2. Theorem 10.1 gives an original Hamilton cycle of their union. Replacing those two factor cycles by this cycle reduces the number of cycles of $\mathcal{F}$, a contradiction. Thus exactly one block, denoted H, is noncograph.

Fix another block D. Every vertex $x \in D$ has a nonempty row A_x into H, since an induced P_4 in H must meet x. Its antineighborhood $H - A_x$ is a cograph. If x, z are consecutive on the specified D-cycle, a spanning H-path between A_x and A_z would close through the spanning D-path obtained by deleting xz, merging two factor cycles. Thus consecutive rows are incompatible. If all rows were strict, Proposition 5.1 and Theorem 5.8 would merge H and D. Some row is therefore saturated. Proposition 5.3, propagated around the D-cycle, makes every row saturated with the same nonempty proper canonical gate S_D. This argument uses

minimality of the factor, without assuming toughness of the two-block induced union.

Suppose satellites D_1, D_2 with different gates have an edge uv, with $u \in D_1$ and $v \in D_2$. Break specified factor edges uu' and vv'. By Proposition 5.3, the different gates supply a spanning H-path between the rows at u' and v'. The two broken satellite cycles become paths from u' to u and from v to v'. Together with uv and the two actual row edges at the ends of the H-path, they form one original cycle spanning H, D_1, D_2. Replacing three factor cycles by this cycle contradicts minimality. Different gate groups are therefore anticomplete.

If there were only one gate group with gate S, every H-satellite edge would have its H-end in S. The tight cut leaves $|S|$ components of $H - S$ and at least one further component among the satellites. Hence $c(G - S) > |S|$ and $c(G - S) \geq 2$, contrary to 1-toughness. There are at least two groups. A vertex of another group is anticomplete to an entire given group. An induced P_4 inside that group would therefore form an induced $P_4 \cup P_1$ with that vertex. Each whole group is a cograph. □

12. Extended cuts on an own group

We extend an old core row across exactly the foreign groups. The old eligible neighbors are checked before locality is applied.

Proposition 12.1 (Cuts on connected own components)**.** *Assume the hypotheses of Proposition 11.1. Fix one canonical group B with old gate S, let R be exactly the union of all other canonical groups, and put*

$$Q = G[V(H) \cup V(R)].$$

Thus H, B, and R partition $V(G)$, and both B and R are nonempty. For $x \in V(B)$, define

$$A_x = N_H(x), \qquad \delta_x = p(Q - A_x) - |A_x|, \qquad T_x = A_x \cup \mathrm{F}(Q - A_x).$$

The graph $Q - A_x$ is a cograph. Every foreign vertex has a neighbor in the original eligible set

$$\mathrm{E}(H - A_x) = V(H) \setminus S.$$

Moreover,

$$\delta_x \geq 0, \qquad c(Q - T_x) = |T_x| + \delta_x.$$

The set T_x lies in Q. It is constant as x ranges over any one connected component of B, and it contains the H-row of every vertex of that component.

Proof. A vertex $x \in B$ has no neighbor in $H - A_x$ by definition. It has no neighbor in R because different gate groups are anticomplete. An induced P_4 in $Q - A_x$ would therefore be isolated from x, so $Q - A_x$ is a cograph.

Let $y \in R$, and write $C_y = N_H(y)$. Both A_x and C_y are saturated rows, but their canonical gates differ. If $C_y \subseteq S = S_{A_x}$, both rows would lie in one tight H-set. Corollary 5.4 identifies their canonical gates. This would make the gate of C_y equal to S, contrary to $y \in R$. Therefore C_y meets $H - S = \mathrm{E}(H - A_x)$. This is an old eligible neighbor in the original core deletion, established separately for every foreign vertex.

Now Theorem 9.1 applies to the fixed graph Q as the foreign cycles are added. If $\delta_x < 0$, one foreign factor cycle witnesses a strict saving. Corollary 8.4 then

assembles one original cycle on H, that foreign block, and the factor block containing x, reducing the factor count. Thus $\delta_x \geq 0$. Proposition 4.9, applied to Q with the cograph remainder $Q - A_x$, gives

$$c(Q - T_x) - |T_x| = p(Q - A_x) - |A_x| = \delta_x.$$

It also makes T_x the unique largest attaining cut containing A_x.

Consider adjacent vertices x, z on one specified own factor cycle. If A_z met an old ineligible vertex released by the foreign extension from $H - A_x$ to $Q - A_x$, Theorem 9.1 would locate either a strict saving or one equal-count release on a foreign factor cycle. In the strict case Corollary 8.4 merges three original factor blocks. In the equal-count case Corollary 8.8 does the same after breaking the specified edge xz of the own cycle. Hence $A_z \subseteq T_x$. Reversing x, z gives $A_x \subseteq T_z$.

Because T_x contains A_z, it is a candidate attaining value δ_x for the maximization defining δ_z; thus $\delta_z \geq \delta_x$. The reverse containment gives $\delta_x \geq \delta_z$. The unique largest attaining-cut property in Proposition 4.9 now gives $T_x = T_z$. Therefore T_x is constant around each own factor cycle.

Next let xz be an actual edge between two distinct own factor cycles, and choose specified factor neighbors x' of x and z' of z. Break the edges xx' and zz'. If $A_{z'}$ escaped $T_{x'}$, locality would give either the strict three-block merger or a spanning $H + D$ path from an actual neighbor of x' to an actual neighbor of z' through one foreign cycle D. In the latter case the two broken own paths and the actual cross edge xz close that path into one original cycle on the four distinct blocks $H, D, \mathrm{block}(x), \mathrm{block}(z)$. Both alternatives contradict minimum factor size. Thus $A_{z'} \subseteq T_{x'}$; the reversed argument gives $A_{x'} \subseteq T_{z'}$. The same scattering comparison and largest-cut argument yield $T_{x'} = T_{z'}$. Constancy around each own cycle propagates this equality across every cross edge in B.

It follows that T_x is constant on each connected component of B. Since every T_y contains A_y, that common set contains all H-rows of the component. □

13. The final group cut

We exclude attachments from nonadjacent own vertices to old endpoints released by foreign cycles. The restriction to old endpoints supplies the nonedges in the forbidden induced graph.

Lemma 13.1 (Nonadjacent rows stay inside the cut)**.** *Under the hypotheses and notation of Proposition 12.1, if $x, z \in B$ are nonadjacent, then*

$$A_z \subseteq T_x.$$

For $z = x$, the containment $A_x \subseteq T_x$ holds by definition.

Proof. Fix nonadjacent $x, z \in B$, put $A = A_x$ and $L = H - A$, and first take one foreign factor cycle D with canonical gate S_D. Let $J = L + D$. Both L and J are cographs: L by Proposition 11.1, and J because x is isolated from its vertex set. Every vertex of D has an actual neighbor in the old set $\mathrm{E}(L)$ by Proposition 12.1. Choose such a neighbor at an eligible endpoint of a minimum cover of L, break the specified cycle of D into a spanning path ending at the adjacent foreign vertex, and append that path. This gives

$$p(J) \leq p(L) = |A|.$$

A strict inequality invokes Corollary 8.4. Hence $p(J) = p(L)$.

Let $r \in \mathrm{F}(L) \cap \mathrm{E}(J)$ be any released old vertex. Theorem 8.7 gives a spanning A-to-r path through $H + D$, both of whose endpoints belong to H. Every H-D edge has its H-end in the old gate S_D, and the tight cut has $c(H - S_D) = |S_D|$. Thus

$$c((H + D) - S_D) = |S_D| + 1.$$

Both endpoints of a spanning path of $H + D$ lie outside S_D. If an endpoint belonged to S_D, deleting S_D from that path would leave at most $|S_D|$ path pieces, fewer than the displayed number of components. In particular, the old endpoint r lies in $H - S_D$ and has no neighbor in D. This conclusion applies to every released old vertex. It makes no assertion about an endpoint lying in D.

Suppose that z sees a released old vertex h. Set

$$C = \mathrm{F}(L) \cap \mathrm{F}(J).$$

Deleting a vertex $f \in \mathrm{F}(K)$ from a cograph K raises $p(K)$ by exactly one. The cut $\mathrm{F}(K) \setminus \{f\}$ gives the lower bound through Proposition 4.5 and Lemma 4.6, and deleting the internal vertex f from a minimum path cover gives the upper bound. An old eligible endpoint stays eligible in a resulting minimum cover. The remaining old ineligible vertices still attain the cut bound. Iterating over C gives

$$\mathrm{E}(L{-}C) = \mathrm{E}(L), \quad \mathrm{E}(J{-}C) = \mathrm{E}(J), \quad \mathrm{F}(L{-}C) = \mathrm{F}(L){\setminus}C, \quad \mathrm{F}(J{-}C) = \mathrm{F}(J){\setminus}C,$$

and both path-cover numbers increase by $|C|$. Every remaining old ineligible vertex is now a released old vertex, so every such vertex misses D.

The vertex h lies in the component of $J - C$ containing D. Outside that component, the old and new graphs have identical components; minimum path covers and endpoint eligibility are componentwise, so eligibility cannot change there. Since h misses D, choose $d \in D$. The connected cograph containing h and d has diameter at most two: the first four vertices of a longer shortest path would induce a P_4. As h and d are nonadjacent, there is an original two-edge path

$$h, u, d.$$

The middle vertex u is old, because h has no neighbor in D. It is not a remaining old ineligible vertex, since all those vertices miss D. Therefore

$$u \in \mathrm{E}(L) = H - S.$$

Now zh, hu, and ud are edges. The pairs zu, zd, and hd are nonedges: $A_z \subseteq S$ while $u \in H - S$; z and d belong to different canonical groups; and h is a released old endpoint that misses D. Hence z, h, u, d induce a P_4. The vertex x misses all four. It misses z by assumption, misses $h, u \in H - A$, and misses $d \in R$ by group anticompleteness. These are five distinct original vertices, so they induce $P_4 \cup P_1$, a contradiction. Thus a nonadjacent z sees no old vertex released by a single equal-count foreign-cycle extension.

Return to the full graph $Q = H + R$. Suppose $A_z \not\subseteq T_x$, and choose $h \in A_z \setminus T_x$. Both old rows lie in $S = A \cup \mathrm{F}(H - A)$, so h is an old vertex of $\mathrm{F}(H - A)$ that is eligible in $Q - A$. Every foreign vertex has an old eligible neighbor, as established before locality in Proposition 12.1. Theorem 9.1 therefore supplies one foreign cycle witnessing either a strict saving or an equal-count release of h. The strict case merges three original blocks. The equal-count case is excluded by the preceding induced-$P_4 \cup P_1$ argument, since zh is an edge and xz is not. Both cases are impossible, proving $A_z \subseteq T_x$. □

Theorem 13.2 (Remaining minimum-factor case)**.** *Let G be a finite simple 1-tough graph of order at least three with no induced $P_4 \cup P_1$. Suppose that G is not Hamiltonian and that a given 2-factor $\mathcal{F}$ has the minimum number of cycles among all 2-factors of G, with at least two cycles. Then the canonical normal form of Proposition 11.1 applies, the extended cuts of Proposition 12.1 satisfy Lemma 13.1, and an original vertex cut contradicts 1-toughness. Consequently no such G and $\mathcal{F}$ exist.*

Proof. Choose a canonical group B, let R be exactly the union of the other groups, and retain the full induced host $Q = G[V(H) \cup V(R)]$ of Proposition 12.1. This choice leaves B and R nonempty and makes H, B, and R a partition of $V(G)$. Fix $x \in B$ and put $T = T_x$.

Proposition 12.1 places in T the H-row of every vertex in the component of B containing x. A vertex z in any other component of B is nonadjacent to x, so Lemma 13.1 gives $A_z \subseteq T$ as well. Thus every H-B attachment has its H-end in the actual vertex set $T \subseteq V(Q)$. There is no B-R edge by the canonical-group separation. Deleting T from the original graph consequently separates the entire intact group B from $Q - T$, and

$$c(G - T) = c(Q - T) + c(B) = |T| + \delta_x + c(B) > |T|.$$

Here $\delta_x \geq 0$, $c(B) \geq 1$, and T contains the nonempty row A_x. The cograph cut identity gives $c(Q - T) = |T| + \delta_x \geq 1$, so $Q - T$ is nonempty. Since B is also nonempty, $G - T$ has at least two components. This is an actual cut of G violating its 1-toughness. □

14. Proof of the Hamiltonicity theorem

The factor-existence and remaining-case results now give the stated cycle without an induction on smaller graphs.

Proof of Theorem 1.1. Theorem 3.1 gives a 2-factor of G. Since G is finite, choose one with the minimum number of cycles. If it has one cycle, that cycle is a Hamilton cycle of G, and every one of its edges is an original edge. If it has at least two cycles and G is not Hamiltonian, Theorem 13.2 gives a cut contradicting 1-toughness. Hence the second case cannot occur in a non-Hamiltonian graph, and G has the required Hamilton cycle. □

References

[KS19] Jinko Kanno and Songling Shan, *Vizing's 2-factor conjecture involving toughness and maximum degree conditions*, Electronic Journal of Combinatorics **26** (2019), no. 2, Paper P2.17. doi:10.37236/7353.

[LWZ26] Yong Lu, Qi Wu, and Qiannan Zhou, *Hamiltonian cycles in 7-tough $(P_4 \cup P_1)$-free graphs*, arXiv:2607.19862v2 (2026). `https://arxiv.org/abs/2607.19862v2`.

[Nik13] Zh. G. Nikoghosyan, *Disconnected forbidden subgraphs, toughness and Hamilton cycles*, ISRN Combinatorics **2013** (2013), Article 673971, 4 pp. doi:10.1155/2013/673971.

[Shan26] Songling Shan, *Hamiltonian cycles in tough $(P_4 \cup P_1)$-free graphs*, Journal of Graph Theory (2026), published online July 1. doi:10.1002/jgt.70090.

[Tut54] W. T. Tutte, *A short proof of the factor theorem for finite graphs*, Canadian Journal of Mathematics **6** (1954), 347–352. doi:10.4153/CJM-1954-033-3.

[Yeo97] Anders Yeo, *A note on alternating cycles in edge-coloured graphs*, Journal of Combinatorial Theory, Series B **69** (1997), no. 2, 222–225. doi:10.1006/jctb.1997.1728.

Computer Science and Technology, Shanghai Jiao Tong University, Shanghai, China